

A qualitative AI security risk assessment of autonomous vehicles

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ARTICLE INFO

Keywords:

AI security
AI in autonomous vehicles
Autonomous vehicles Security
RL security

ABSTRACT

This paper systematically analyzes the security risks associated with artificial intelligence (AI) components in autonomous vehicles (AVs). Given the increasing reliance on AI for various AV functions, from perception to control, the potential for security breaches presents a significant challenge. We focus on AI security, including attacks like adversarial examples, backdoors, privacy breaches and unauthorized model replication, reviewing over 170 papers. To evaluate the practical implications of such vulnerabilities we introduce qualitative measures for assessing the exposure and severity of potential attacks. Our findings highlight a critical need for more realistic security evaluations and a balanced focus on various sensors, learning paradigms, threat models, and studied attacks. We also pinpoint areas requiring more research, such as the study of training time attacks, transferability, system-based studies and development of effective defenses. By also outlining implications for the automotive industry and policymakers, we not only advance the understanding of AI security risks in AVs, but contribute to the development of safer and more reliable autonomous driving technologies.

Contents

1.	Introduction	2
2.	Background	3
2.1.	Artificial intelligence in AVs.....	3
2.2.	AI security	4
2.2.1.	Threat modeling AI	4
2.2.2.	Attacks on AI.....	5
2.3.	AV AI security	6
3.	Methodology	6
3.1.	Measuring attack exposure	7
3.2.	Measuring threat severity	7
3.3.	Literature analysis	7
4.	AV AI component security risk	8
4.1.	Perception	8
4.1.1.	Traffic sign recognition.....	8
4.1.2.	Action recognition.....	9
4.1.3.	Object detection.....	10
4.1.4.	Object tracking	11
4.1.5.	Image segmentation.....	13
4.1.6.	Depth estimation.....	13
4.1.7.	Optical flow.....	15

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4.1.8.	Camera pose estimation	15
4.2.	Prediction	16
4.2.1.	Trajectory prediction	16
4.3.	Planning	16
4.4.	Control or lane keeping	17
4.5.	End-to-end driving	19
5.	State of research, challenges, and future research directions	19
5.1.	Landscape of attack evaluations	20
5.2.	AV system security evaluations	21
5.3.	Real-world or problem-space security evaluations	21
6.	Implications beyond research	22
6.1.	Implications for the automotive industry	22
6.2.	Implications for policymakers	23
7.	Concluding remarks	23
	CRediT authorship contribution statement	23
	Acknowledgments	23
	Appendix. Object detection overview table	23
	References	24

1. Introduction

Artificial intelligence (AI) security has been studied extensively (Gu et al., 2017; Serban et al., 2020; Li et al., 2021b; Akhtar et al., 2021; Cinà et al., 2023; Liang et al., 2022). An attacker can exploit AI models in different ways. An AI's integrity can be compromised when specific training inputs force the model to wrongly output an attacker-chosen result at test time due to a backdoor (Gu et al., 2017). Integrity can also be harmed at test-time by carefully perturbing the data (Akhtar et al., 2021). In addition, the attacker can tamper with the training data to hurt availability or the overall performance of the model (Cinà et al., 2023). Sponge attacks increase the time the model requires to process the given test input (Shapira et al., 2023). Finally, at test-time, an adversary can copy a provided AI model without the owner's consent (Liang et al., 2022) or extract private data from it (He et al., 2020), violating confidentiality (Liang et al., 2022). By customizing attacks, security issues arise almost regardless of the specific application or model details (Akhtar et al., 2021; Serban et al., 2020; He et al., 2020; Yu and Liu, 2022), and many test time attacks require only presenting specific, adversarial inputs to the sensor that records the AI's input.

AI security breaches are problematic with AI's growing adoption, which has also recently been acknowledged in legislation. For example, the EU AI Act requires secure AI, with a penalty of at least 7.5 Million Euros or 1.5% of annual turnover for non-compliance.¹ Hence, there is substantial interest in the topic. There are surveys about AI security in image classification (Akhtar et al., 2021) or object detection (Serban et al., 2020). Other surveys focus on the physical feasibility of attacks (Sun et al., 2018), attacks at training time (Cinà et al., 2023), privacy (Hu et al., 2022) or stealing proprietary AI models (Oliyynyk et al., 2023). While such a systematization of knowledge is crucial, less work considers the AI security risks of a system composed of several AI-based components. We targeted this gap and chose a single system composed of potentially many AI tasks, autonomous vehicles (AVs).

Non-autonomous vehicles are involved in many accidents, >90% of which are caused by human error.² AVs have thus an enormous potential to save lives—under the condition that they operate reliably with few errors. Recent events question this reliability (Grosse et al., 2024b).^{3,4} In addition, street-legal vehicles with AI components have been shown to misclassify traffic signs (Povolny and Trivedi, 2020) or to fail to detect street lanes (Sato et al., 2021; Jing et al., 2021) when attacked. Either will strongly affect their behavior to cruise and keep their passengers safe. Researchers have demonstrated more attacks, however not in the physical or problem space, but in simulations. Effects on the vehicle are to disregard people with specific clothes (Ma et al., 2022; Wei et al., 2023), to ignore objects in general (Hu et al., 2021; Wu et al., 2023a), to hallucinate objects which are not there (Song et al., 2018; Zhu et al., 2023a), or to neglect the depth of an object (Daimo and Ono, 2023; Guesmi et al., 2023). It has also been shown that the internally predicted trajectory of other road users can be tampered with Tan et al. (2023) and Saadatnejad et al. (2022), or the perceived actions of pedestrians arbitrarily altered (Pony et al., 2021; Hammoud et al., 2023). Some AI-based systems within the vehicle can also be caused to time out (Vemprala and Kapoor, 2021; Shapira et al., 2023) and become unresponsive. Each AV may thus be potentially vulnerable to overlooking or hallucinating objects or to becoming unresponsive, severely threatening the safety of passengers or any road user.

Most attacks are tested in simulation. Yet, some attacks were shown in the problem space: by, for example, printing a perturbation (Lu et al., 2017; Yamanaka et al., 2020) or using objects like traffic cones (Han et al., 2022), capturing them via camera, thereby feeding them into the AI-based component. In addition, few existing approaches test how these attacks impact the entire vehicle system (Morgulis et al., 2019; Jing et al., 2021; Cao et al., 2021; Hallyburton et al., 2022; Cao et al., 2022; Koren

¹ https://www.europarl.europa.eu/pdfs/news/expert/2023/12/press_release/20231206IPR15699/20231206IPR15699_en.pdf.

² <https://www.cbmc1aw.com/what-percentage-of-car-accidents-are-caused-by-human-error/>.

³ www.foxnews.com/auto/tesla-smashes-overturned-truck-autopilot.

⁴ <https://www.ktvu.com/news/self-driving-car-blocks-police-responding-to-san-francisco-shooting>.

et al., 2022). This aligns with general claims that AI security research does not necessarily apply to real-world settings (Bieringer et al., 2022; Apruzzese et al., 2022; Cinà et al., 2023; Grosse et al., 2023). Our goal for this review is thus two-fold. Firstly, we review *theoretically possible attacks on AVs*. Secondly, and orthogonally, we analyze how *realistic* the found attacks are, in other words, how likely they can be executed in the problem space on an existing, street-legal vehicle.

Contributions. Our goal is to analyze the security risk for AI within an AV. In the first step, we identify possible affected system parts (AV pillars in Section 2 and AI components in Section 2.1). We then describe the existing threats (Section 2.2). After these first steps, we can describe the risk as $\text{risk} = \text{exposure} \times \text{severity}$ (Bayne, 2002). Bayne (2002) defines *exposure* as related to the attack surface, whereas the latter *severity* are the resources needed to exploit the attack. Both of these quantities are, in practice or on the road, open research questions (Grosse et al., 2024a). We thus instead propose a qualitative analysis, where we define measures for exposure and severity (Section 3) and use existing literature to analyze the proposed risk measure (Section 3.3). The analysis of each component reveals shortcomings and needed future work in each field (Section 4). We then discuss research challenges and future work (Section 5). On the attack level, for example, there is a lack of approach diversity and proper evaluations. Beyond the attacks, the AV poses challenges when studied as a whole system, which should consider a diversity of sensors and approaches, but be inspired by existing AVs. Finally, studying the real-world security of AVs requires a better understanding of the existing threat surface and studies of the corresponding settings. Due to the ubiquity of vehicles, our work has implications beyond research (Section 6). For industry, we outline the need to consider security, security's benefits in improving safety, and the need for collaborations between industry and academia to study relevant AV systems. For policymakers, we sketch the need to update current legislation, build new specialized units, and collect occurring incidents.

Related work. Several surveys also cover the AI security of AVs. However, none of these focuses on the real-world feasibility of attacks. We will now review their contribution and differences to our paper. Girdhar et al. (2023) focus on the cybersecurity of AVs and discuss AI attacks and defenses only briefly. Many surveys do, however, focus on AI security. For example Wei et al. (2022) discuss AI security of AVs, but do not discuss, in contrast to our work, the task specifics of the attacks. While Shen et al. (2022) consider these components, they analyze AV AI security to propose a platform to test AV system-level attacks. Deng et al. (2021) do not, in contrast to us, consider any threat model when reviewing the attacks, and consider orthogonally to us attacks on the sensor level. Furthermore, Demontis et al. (2022) cover RL for AVs: their work discusses many attacks on RL tasks that could occur within AVs, but do not target AVs directly. Their survey complements our analysis of AI paradigm-independent attacks on all AV components. Sun et al. (2018) focus on a case study with only three attacks, only one of which focuses on an AV-relevant task, traffic sign recognition. While we examine more than twice as many papers as these previous surveys, there are more differences. The works by Wei et al. (2022), Shen et al. (2022), and Deng et al. (2021) only cover papers up to 2021, not up to 2024 like us. They do not, for example, report attacks on planning or lane keeping (Shen et al., 2022). Finally, most other works deal also with defenses (Demontis et al., 2022; Wei et al., 2022; Shen et al., 2022; Deng et al., 2021). We focus instead on attacks to allow an in-depth treatment of the evaluations in the problem space (exposure) and needed resources (severity). This focus also allows us to identify shortcomings in terms problem-space security for each task within the AV.

2. Background

We first review the relevant background around autonomous vehicles (AVs), describing on which sensors AVs rely and which AI components are used within the AV. We then discuss artificial intelligence (AI) with the relevant paradigms for this paper and conclude with AI security.

Autonomous vehicles. AI within AVs is structured using four pillars (Sadat et al., 2020; Saadatnejad et al., 2022; Deng et al., 2021) which we visualized in Fig. 1. The first pillar, perception, relies on an array of sensors (Georgia et al., 2021). Here, we review the ones covered in this analysis. These sensors include cameras and LiDAR, a radar variant, which is represented as point clouds. Often, not only a single sensor is used, but several sensors are fused for more reliance (Cao et al., 2021; Hallyburton et al., 2022). These sensors are the basis for the first pillar of the AV, perception. Perception encompasses tasks like traffic sign classification (Gu et al., 2017), action recognition (Mordan et al., 2021), object detection (Zhu et al., 2023a), object tracking (Chen et al., 2020), image segmentation (Nakka and Salzmann, 2020), depth estimation (Yamanaka et al., 2020), optical flow (Ranjan et al., 2019) and pose estimation (Inkawhich et al., 2018). Given an understanding of the environment based on these methods, the prediction module models, for example, pedestrians (Cao et al., 2022; Hwang et al., 2021) or other vehicles (Zhang et al., 2022a). Next comes planning (Ma et al., 2019) to determine the exact path the vehicle will take. In the last pillar, control, the subsequent actions are executed (Han et al., 2022). As an alternative to the pillars, we can model a vehicle end-to-end (Yu and Liu, 2022). Each task, or end-to-end modeling, encompasses vulnerabilities, where attack details, consequences, and evaluations depend on task specifics.

2.1. Artificial intelligence in AVs

AI encompasses a variety of algorithms that can *learn* from data, in the sense that they do not contain hard-coded instructions, but rather learn the task from the data provided. Several paradigms exist in AI, of which we review two for our analysis: machine learning (ML) and reinforcement learning (RL).

Machine learning in AVs. In ML, a task consists of, for example, outputting for an image whether it contains a *stop* or a *yield* sign. More concretely, the data or images are $\mathcal{X} \subset \mathbb{R}^n$, and the labels $\mathcal{Y} \subset \mathbb{N}$ represent the class assignment. Taking some training data from the original distribution, e.g. $X \in \mathcal{X}, Y \in \mathcal{Y}$, we learn the classifier on this data. In other words, we fit a mathematical function f by minimizing the classifier error or loss on this training data $\{X, Y\}$, e.g. $\min_{\omega} \mathcal{L}(f(\omega, X), Y)$. An instance of such a loss

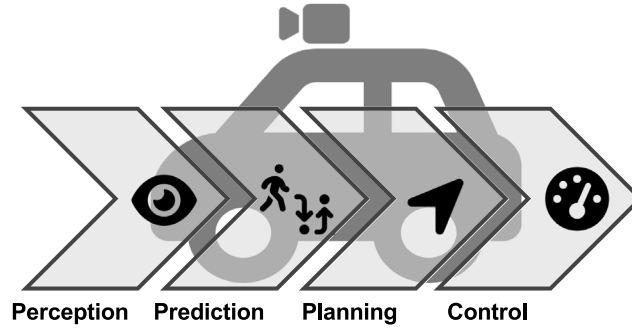


Fig. 1. The four pillars of an autonomous vehicle (Sadat et al., 2020; Saadatnejad et al., 2022).

Table 1

Attack taxonomy, as introduced by Biggio and Roli (2018), with updated references. We distinguish the attacker's capability in accessing training or test data. There are three goals the attacker can pursue, causing individual, wrong outputs (integrity), decreasing overall performance (availability), and circumventing the secrecy of data or model (confidentiality).

Attacker's capability	Attacker's goal		
	Integrity	Availability	Confidentiality
Training data	Backdoors (Ma et al., 2022)	Poisoning (Ma et al., 2019)	–
Test data	Evasion (Yamanaka et al., 2020)	Sponge (Shapira et al., 2023)	Model stealing (Liang et al., 2022)

function $\mathcal{L}$ is the cross-entropy loss. After the training is finished, the classifier generalizes and classifies the unseen test data X_t correctly, e.g., $f(\omega, X_t) \approx Y_t$. ML encompasses a range of output distributions and corresponding loss functions: regression predicts a real number, thus $\mathcal{Y} \in \mathbb{R}$. In image segmentation and other pixel-wise prediction tasks, $\mathcal{Y} \in \mathbb{R}^n$, the output has the same shape as the input. In object detection, bounding boxes (e.g., locations of objects) are predicted.

Reinforcement learning in AVs. An example of RL in an AV is navigation in a chess board-like environment. The agent, $\mathcal{F}$, interacts with the environment $\mathcal{E}$ (the chessboard) in terms of a set of predefined actions $\alpha_1 \dots \alpha_n$ (move up/right/left/down). These actions are determined by the policy π applied to the observation o (ability to move on the board) of the current state s (the position) of the environment $\mathcal{E}$. Formally, $\alpha_t = \pi(o_t(s_t(\mathcal{E})))$, where an action yields a new state and observation, e.g. $o_{t+1}, s_{t+1} = \alpha_t(s_t, \mathcal{E})$. Each state s_t is associated with a reward R , and analogous to minimizing the loss in ML, the reward R is maximized in RL, e.g. $\max_{\pi} \sum_t \gamma^t R(s_t)$. We here omit for simplicity that the state depends on the previous state, action, observation; and thus the policy. This policy can be explicitly learned, or implicitly represented, and many different approaches and strategies exist to solving this problem. We here refer the interested reader to the surveys by Kaelbling et al. (1996) and Sewak (2019).

2.2. AI security

Having revisited the basics of AI, we are ready to discuss AI security. Security studies measures to prevent attacks on systems. To this end, we first model AI in terms of possible security threats. This threat model defines the attacker and is reviewed first. Afterwards, we detail the individual attacks and their applicability in the problem space to an AV.

2.2.1. Threat modeling AI

To describe the attacker, a threat model is used. We follow the threat model by Biggio and Roli (2018), who use the dimensions of goal, knowledge, and capabilities to characterize the attacker.

Attacker's goal. In a nutshell, the attacker may cause three violations (Biggio and Roli, 2018): The first one, *integrity*, preserves the benign performance of the algorithm but leads to individual, wrong outputs. The second, *availability*, decreases overall performance. Finally, the third, *confidentiality*, concerns the training data's secrecy and the intellectual property of the model. Attacks usually achieve one of these goals, as visible in Table 1. The exact implementation of these high-level goals is highly task-specific. On the one hand, an availability violation in object detection may mean that the algorithm outputs a large number of bounding boxes (Shapira et al., 2023). On the other hand, a planning algorithm may take longer to compute a solution (Vemprala and Kapoor, 2021). As these outputs are likely used within the task, it is hard to assess the concrete impact of an individual attack on an AV. This is thus left for future work. We instead discuss the attacker's goal for each task separately.

Attacker's knowledge. The attacker's knowledge entails the classifier f and the applied algorithm or the weights ω or the policy π in RL. We add in this work how much knowledge the attacker has about the task. More specifically, we refer to knowledge like the input being encoded as RGB matrices. We denote this $\mathcal{D}$, e.g., distributional knowledge. For each of these properties, knowledge can either be present (white-box, denoted $\circ$) or absent (black-box, $\bullet$), the latter meaning that the attacker does not know.

Attacker's capabilities. The capabilities encompass whether the attacker can query the model f , alter the training data X , the training labels Y , or the test data X_t . Each capability is visualized as executable ($\circ$) or not ($\bullet$). The exact changes performed on

the data depend on the desired goal and are based on the attack, which we discuss next. Changes can be applied to labels or to the data samples. When applied to samples, they are generally denoted as perturbation δ , and can be additive, based on a function, or even more complex when changes occur in the problem space, hence before being captured. In the following, we will always write additive perturbations for simplicity and without loss of generality.

2.2.2. Attacks on AI

An attack achieves a specific goal of the attacker. We depict the attacks with their associated goal in Table 1. This table contains a second, AI-specific distinction: whether an attack is executed during or after training. This latter distinction is highly relevant for settings like AVs, where the deployed system in the car is easily accessible at test time, but the training is done within the company. In the latter case, at training time, there is a higher cost for the attacker to access the system or tamper with the data.

We start our description with backdoors and poisoning, two training time attacks, and then continue with test time attacks achieving various goals.

Backdoors. The attacker aims to harm integrity by implanting a backdoor in the model. This backdoor consists of a specific perturbation or trigger causing the model to output the attacker's desired class or behavior (Cinà et al., 2023; Messaoud et al., 2023; Nguyen and Tran, 2021). This trigger or perturbation δ can be pixels (Lan et al., 2024) up to real objects like clothes (Ma et al., 2022). In general and simplified, the attacker builds a new dataset from the original data $\{X, Y\}$ of the form $\{X, Y\} \cup \{X' + \delta, Y'\}$, where the second term denotes the data with the trigger (δ) and the target labels (Y'). The procedure is somewhat analogous in RL, where the trigger is embedded in the agent's observations o (Mo et al., 2022), or behaviors of another agent within the environment $\mathcal{E}$ (Gong et al., 2022). Analogously to the training labels, in RL, the reward $\mathcal{R}$ is altered (Gong et al., 2022) or knowledge about it exploited (Mo et al., 2022). In all cases, after training, the backdoor is exploitable (Cinà et al., 2023; Messaoud et al., 2023; Nguyen and Tran, 2021; Lan et al., 2024; Mo et al., 2022). In other words, whenever the trigger is present, the target class is outputted or $f(X_i + \delta) = Y'_i$ or the target action(s) performed. To summarize, backdoor attacks require at least access to training data (and labels) and test inputs. In some cases, the attacker requires additional access to or knowledge about the model's training procedure. Numerous defenses against backdoors have been proposed (Cinà et al., 2023). There is an ongoing arms race (Shokri et al., 2020; Cinà et al., 2023), where increasingly strong attacks break defenses. According to Cinà et al. (2023), there are six strategies to tackle backdoor attacks: sanitizing the data before training, implementing more robust training to backdoors, or defenses at test time. These include scanning the model for backdoors, reconstructing triggers, removing a trigger from a model, or sanitizing the incoming test data.

Poisoning. In poisoning attacks, the attacker alters the training data or labels to decrease the accuracy of the resulting classifier. To this end, training samples (Mei and Zhu, 2015), labels (Biggio et al., 2011), or both (Mo et al., 2022) are altered. To understand how poisoning works, we first revisit the common training procedure to obtain weights ω to minimize the loss $\min_{\omega} \mathcal{L}(f(\omega, X), Y)$. The attacker now perturbs the data to maximize the loss during the victim's training. In other words, the attacker implements a bilevel strategy to, for example, alter the data to decrease the victim's performance $\max_{\delta} \min_{\omega} \mathcal{L}(f(\omega, X + \delta), Y)$. Here, the outer maximization is the attacker's goal and the inner minimization represents the victim's training. Although superficially different, the same problem is solved in RL, where a suboptimal policy π is learned due to a tampered-with reward $\mathcal{R}$ (Xu et al., 2021), written $\min_{\delta} \max_{\pi} \sum_i \mathcal{R}_{\delta}(s_i)$. Analogously to ML, the changes to the rewards can also be determined using bilevel optimization (Xu et al., 2021) or RL (Zhang et al., 2020a). To execute a poisoning attack, knowledge of the training data and sometimes the model is required, and the training data must be accessible. Poisoning is, in terms of defenses, rather well understood (Cinà et al., 2023). For example, there is a trade-off between attack strength and detectability (Frederickson et al., 2018) and regularization helps to mitigate attacks (Demontis et al., 2019).

Evasion. Evasion or adversarial examples harm the model's integrity: a small perturbation δ is added to inputs at test time, causing the model to compute a wrong output (Dalvi et al., 2004). In general, the attacker applies a minimal perturbation δ to a test sample x , causing the wrong output: $\min_{\delta} f(\omega, x + \delta) \neq f(\omega, x)$. This equation is similar to the training described in Section 2.1, which does not alter the weights ω but the input sample. The formulation can be altered to yield a specific model output Y_i^* , for example, outputting a specific class (Benz et al., 2020). The latter is called *targeted*, the first *untargeted* attack. Many approaches have been suggested to solve this approach. The fast gradient sign method takes one step in the gradient's direction from the optimization problem above (Szegedy et al., 2014; Inkawhich et al., 2018). Another approach is projected gradient descent (Cao et al., 2021; Madry et al., 2018), but attacks without gradient computations are also possible (Yan et al., 2022; Jiang et al., 2019). The overall approach works analogously for RL, with the difference that there are more possibilities to apply a perturbation δ . The attacker can here optimize for a perturbation δ applied to state s (Ren et al., 2023), observation o (Sun et al., 2020) or environment $\mathcal{E}$ (Boloor et al., 2020). The problem formulation is however analogous, minimizing the reward $\min_{\delta} \sum_i \mathcal{R}_{\delta}(s_i)$ for untargeted attacks. To summarize, evasion requires access to the model's test inputs and ideally also to the model. Analogous to backdoors, there is an ongoing arms race (Carlini and Wagner, 2017; Carlini et al., 2019), where many proposed defenses are broken. A possible defense to increase the attacker's cost is to train the model on adversarial examples, also called adversarial training (Croce et al., 2021). An alternative is formal verification, which is however not applicable to all sizes of models and all domains (Meng et al., 2022).

Sponge attacks. Sponge attacks harm the availability of the model at test time. For example, the model's output is made unusable (Shapira et al., 2023) or the model's outputs take more time (Vemprala and Kapoor, 2021). To the best of our knowledge, no sponge attacks have been introduced on RL. Sponge attacks only require access to the test samples. They were recently introduced by Shumailov et al. (2021) and are not well understood in terms of defenses.

Model stealing. Model stealing harms the confidentiality of the model at test time: the attacker copies the ML model functionality without the owner's consent (Tramèr et al., 2016). The attacker, given black-box access to the original model f , tries to reproduce

a model with similar performance, e.g. $f'(X_t) = Y_t \approx f(X_t) = Y_t$. This is for example done by using the label outputted by the target classifier, e.g., $f(X_t)$. As Oliynyk et al. (2023) discuss, there are different constraints on how close the performance and the exact layout and weights ω should match. Model stealing has been demonstrated on RL (Chen et al., 2021a), but not in an automotive context. To summarize, for model stealing, test samples and the corresponding target model's outputs are needed. As for the previous attacks, mitigations have been proposed (Juuti et al., 2019; Orekondy et al., 2020), of which few have been broken (Oliynyk et al., 2023). Beyond denying access to outputs, it is thus an open question of how to defend model stealing. Oliynyk et al. (2023) discuss defenses like model identifiers, watermarks, and monitors.

Privacy. Several attacks that harm the dataset's confidentiality or privacy. Here, we are concerned with membership inference (Shokri et al., 2017; Pan et al., 2019). The goal is to determine, given the model's outputs $f(x)$, for a sample x whether it formed part of the training data, e.g. $x \in X$. Analogously, in RL, the training environments can be deduced (Pan et al., 2019). Summarizing, to harm the privacy of a model, the attacker needs access to input and output pairs, analogous to the previous attack, model stealing. Defenses have been proposed, of which some like distillation have been broken (Jagielski et al., 2024). It remains thus an open problem how to reliably defend against membership inference.

2.3. AV AI security

To conclude this section, we briefly elaborate on how an AV and its AI components are vulnerable. Summarizing the previous subsections, there are two main ways to attack any AI component. One is via the training data and thus the model (Ma et al., 2019, 2022). Whether this is feasible in practice is an open research question (Grosse et al., 2024a). Secondly, the attacker can use the test inputs (Yamanaka et al., 2020; Liang et al., 2022). Each time the AV processes data from its surroundings, for example via a camera or LiDAR, these inputs can potentially be altered maliciously and influence individual components.

Still, the attentive reader may be skeptical that an evasive perturbation on, for example, a digital model may apply to a real-world AV. We will now discuss several AI security community findings showing that even an unknown model in a prior unknown scene can be fooled reliably.

Proprietary models. Except backdoors, most attacks require access to the target model for, for example, gradient computation. This seems a strong drawback when targeting a largely proprietary AV. On the one hand, having access to model outputs may suffice to approximate the necessary gradients (Kuang et al., 2021). Such attacks are called black-box, according to the attacker's knowledge from their threat model. On the other hand, *transferability* enables to compute attack samples x_a on one model or agent, f^1 , and use the obtained samples to target a second model or agent $f^2 \neq f^1$ successfully (Huang et al., 2017; Demontis et al., 2019). Transferability has been shown for evasion and poisoning in ML (Demontis et al., 2019) and evasion in RL (Huang et al., 2017) and implies that also proprietary models are vulnerable.

Unknown model inputs. Formulations like evasion derive a perturbation δ for a specific sample x (Kuang et al., 2021). For an AV, the digital input is unlikely to be accessible to the attacker. Knowing which sensor is likely to be used and the task at hand is however often enough (Morgulis et al., 2019). In addition, the evasion formulation can be altered to work on several samples or a batch (Thys et al., 2019), becoming *universal*. In other words, the perturbation is more likely to work on samples not used during the attack computation (Thys et al., 2019). However, to the best of our knowledge, no work shows universal attacks on RL. Other attacks, like backdoors, are universal on ML by design: the trigger will, after training, be effective on any sample (Gu et al., 2017; Yi et al., 2023). This is not the case for RL agents (Yang et al., 2020a), where the trigger may be less reliable (Mo et al., 2022).

Implementable perturbations. In the discussion so far, we have skipped how perturbations are measured. Due to their practicability within optimization frameworks, often, L_p -norms are used (Tian et al., 2018; Xu et al., 2020; Gu et al., 2022; Wu et al., 2023b; Jiang et al., 2023). In practice, implementing a perturbation with a small L_∞ -norm may be impossible, as this would mean perturbing every input pixel. Perturbations can however be contained to patches, which when applied to objects could affect an AV (Ranjan et al., 2019; Wortman, 2021; Nesti et al., 2022; Schrodi et al., 2022). This requires however that the perturbation δ maintains its malicious effect when printed or deployed in the problem space or real world, which is the case (Sitawarin et al., 2018). In addition, the attacker may have less control over the distance or the angle the perturbation is captured from. Hence, Athalye et al. (2018) incorporate different functions like rotation, scaling, and translations into the loss to obtain the perturbation δ . We depict examples of digital and real-world perturbations in Sections 4.1.1 and 4.1.3 in Figs. 2 and 3.

3. Methodology

Given that AVs are potentially vulnerable, we define the terminology and measures to quantify AV AI security risk in a qualitative approach. As stated in the introduction, risk in a cybersecurity context can be assessed as $risk = exposure \times severity$ (Bayne, 2002). This assessment ideally entails rating the implications or impact of possible attacks (Bayne, 2002). Given the absence of real-world AI security incidents (Grosse et al., 2024b) and thus real-world impact, a possible approximation would be whether an attack is targeted (e.g., to what degree the attacker can control the model's output). However, targeted attacks have lower success rates in more realistic black-box settings (Naseer et al., 2021). We thus leave an in-depth analysis of the attack's impact to future work, and define only measures that approximate exposure and severity. Our approach is qualitative due to the lack of documented AI attacks on AVs (Grosse et al., 2023), and we thus conclude this section with a description of our literature search.

Table 2

Definition of exposure (E). We consider the test environment and how many constraints are incorporated, where ++ denotes complex constraints and + simple constraints like for example expectation over transformation (Athalye et al., 2018).

Test environment	Constraints incorporated		
	++	+	None
Production vehicle	●	●	●
Production components ^a	●	●	●
In the problem space	●	●	●
Otherwise/only digitally	○	○	○

^a Like Baidu Apollo.

3.1. Measuring attack exposure

Given that real exposure or attack surface is an open research question (Grosse et al., 2023), we opt for an approximation: We propose to measure exposure E by quantifying how real-world like the evaluation of the article was. We reason that an attack with a more practical evaluation is more likely to show the attack surface of a real AV. An attack tested on a street-legal AV should thus obtain the highest score of E, whereas an attack in the digital space on an isolated model should receive the lowest score.

We also take into account physical constraints when mounting the attack. One category (+) incorporates small changes to the loss. An example is expectation over transformation (EoT), which formalizes distortions like rotation, scaling, and translation (Athalye et al., 2018), as discussed in the previous section. The second category (++) contains more complex constraints, like mimicking the style of existing objects (Zhu et al., 2023a), or using real objects/movements as perturbations (Han et al., 2022; Zhang et al., 2022a). We extend the informal notion from previous work (Llorca et al., 2024), and define measure E as depicted in Table 2. The criteria we use can be easily deduced from a given research paper, as in the following metric for severity.

3.2. Measuring threat severity

The second factor, severity, is defined by Bayne (2002) concerning the required resources. We thus define a severity measure S that quantifies how few resources are needed to exploit a vulnerability. We reason that attacks relying on fewer resources are easier to mount and thus potentially more dangerous to an AV. As we rely on a qualitative analysis, each resource is deducible from and depicted for a given attack paper.

- **Query model access.** The attack requires no knowledge about the target model f 's weights (●). Instead, the gradients are approximated by querying the model.
- **No model access.** No access at all to the model is required: the attack transfers (*transf.*, ✓): it has been shown to affect a second model other than the one it crafted for.
- **Executability.** The perturbation is executable over time (*exe.*, ✓). This excludes sequences where perturbations differ strongly across frames, but includes constant perturbations or, for example, drivable maneuvers. A special case is perturbations optimized to be effective from different angles (✓*).
- **Universality.** The attack does not require a specific sample: perturbations generalize across different inputs (*univ.*, ✓). Non-universal perturbations are specific to one input or scene and are ineffective outside this setting or scene.
- **Perturbation style.** An attack is not global and does not require changing the entire sample (δ , task-dependent). Otherwise, the attacker would have to perturb the entire scene, including pedestrians, other vehicles, and the sky. Examples for feasible perturbations include perturbations that resemble stickers and objects or are for example other agents' trajectories.

Each of the points model access, universality, perturbation style, and time independence count as 25%, whereas query model access counts as 12.5%. In other words, if all of the above resources are needed, S is 0% or ○, and with each full resource not necessary increases by 25%, the highest severity being ●. A query-based, universal attack with a sequential global perturbation (hence time-dependent) receives thus ○. The most severe attack (●) can be mounted without knowledge of the target model, works on any sample that is only locally changed, and this single perturbation suffices for attack success. Our measure S is a lower bound, as an attack may transfer despite missing evidence, or as a global perturbation may indeed be implementable (like flickering (Pony et al., 2021), or blur (Ji et al., 2021; Zhu et al., 2023a)).

3.3. Literature analysis

Both of the previous metrics can be derived from a given research paper, making it crucial to analyze all papers available for a given pillar and task. We thus briefly describe how we searched for the papers that we analyzed. Using Google Scholar, we used terms like for example "action recognition", "security", and "backdoors" and searched top-tier conferences (Usenix Security, IEEE Security & Privacy Symposium, NeurIPS, ICCV, etc.), as well as within cited and citing papers. As we found hundreds of papers, we narrowed down the scope of the analysis: The article must contain a task directly related to autonomous driving (e.g., traffic light detection),

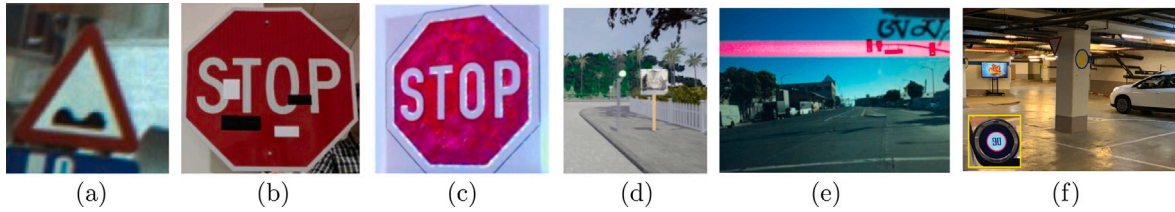


Fig. 2. Different (implementations of) perturbations δ on image data. (a) A global perturbation spans the entire image (L_∞) (Jiang et al., 2020). (b) A local, patch-based perturbation (P) (Eykholt et al., 2018). (c) An object-shaped perturbation (O) is projected onto the target (Worzyk et al., 2019). (d) A billboard contains an adversarial perturbation (O) rendered in a scene (Patel et al., 2020). (e) A Laser beam is a perturbation (O) (Hu et al., 2024). (f) Experiments on a real vehicle using a screen depicting a billboard with a perturbation (O) (Nassi et al., 2020).

not only a related task (e.g., image classification). In fields with many publications (object detection, image segmentation), we were stricter about this constraint than when few publications were available (e.g., path planning, action recognition). We exclude attacks that do not explicitly target AI but sensors (Yan et al., 2016), sensor output (Köhler et al., 2021), or other hardware components (Sun et al., 2023) to limit the scope of this review. Nonetheless, we found 177 papers from early 2017 to early 2024.

We excluded defenses, as acceptably covering defenses for each task would require discussing the arms race (e.g., broken defenses). Attacks breaking defenses may not necessarily be from an automotive context, and we thus leave systematizing the state of defenses for AI in AVs for future work.

4. AV AI component security risk

To analyze the qualitative risk of AI security threats on AVs, we rely on the four pillars (Sadat et al., 2020; Saadatnejad et al., 2022) as a scaffold and review perception, prediction, planning, and control. For each of these pillars, we discuss all individual tasks and how each task is modeled in terms of AI and AI security. For the latter, we discuss the attacker's goal, how the attacker perturbs the data, and the perturbation's physical constraints. Afterwards, we review attack exposure and severity, assess the qualitative risk, and draw a conclusion.

4.1. Perception

Perception contains a plethora of tasks. We start with traffic sign recognition and action recognition, two classification tasks, and then continue with object detection and object tracking. Afterwards, we discuss pixel-wise prediction tasks like image segmentation, depth estimation, and optical flow. We conclude with the regression task of (camera) pose estimation. All tasks serve the AV to *perceive* its surroundings. In some areas, only a single paper has been published so far. These include bird's eye view analysis (Zhu et al., 2023b), graph scene generation (Zhao et al., 2024), human pose estimation (Duan et al., 2023), and 3D human pose estimation (Chen et al., 2021b). We thus do not discuss these areas in depth.

4.1.1. Traffic sign recognition

Traffic sign recognition is an image classification task, where we classify an image of a stop sign as containing the former (Sitawarin et al., 2018; Zhong et al., 2022; Liu et al., 2019; Dong et al., 2022). A special case is traffic light recognition, where the desired output is “green”, “orange”, or “red” (Yan et al., 2022; Patel et al., 2020). Due to a large number of published papers, we focus on all vehicle-specific articles (as depicted in Table 3) and refer the interested reader to surveys in image classification security (Akhtar et al., 2021) or training time attacks thereon (Cinà et al., 2023).

Attacker's goal. Two training time integrity, or backdoor, attacks cause attacker-defined outputs at test time (Nguyen and Tran, 2021; Gu et al., 2017; Patel et al., 2020). When untargeted, integrity at test time or evasion causes wrong output (Jiang et al., 2020; Sitawarin et al., 2018; Zhong et al., 2022; Gnanasambandam et al., 2021). Targeted attacks lead to specific output classes and classify a “stop” sign as “yield” to cause an accident (Worzyk et al., 2019; Liu et al., 2019; Gnanasambandam et al., 2021). In traffic light classification, targeted attacks swap “green” to “red” and vice versa (✓*) (Yan et al., 2022). One availability attack also decreases the model's performance during training (Jiang et al., 2020).

Perturbations. We give illustrations of frequent perturbation classes from the papers below in Fig. 2. Few papers use global perturbations (L_∞ , Fig. 2(a)), e.g., pixel changes over the whole image to decrease accuracy at test (Jiang et al., 2020; Benz et al., 2020; Sayles et al., 2021; Gnanasambandam et al., 2021) or at training time (Jiang et al., 2020; Yi et al., 2023). In contrast, changes can be constrained to smaller areas or patches (P, Fig. 2(b)) (Zhong et al., 2022; Benz et al., 2020; Bai et al., 2021). Another class of perturbations is object-shaped. Such perturbations are rendered or projected on objects (O, Fig. 2(c)–(e)) (Liu et al., 2019; Duan et al., 2020; Worzyk et al., 2019), or have the shape of a laser beam (Duan et al., 2020; Yan et al., 2022; Yufeng et al., 2023; Hu et al., 2024), or a billboard (Patel et al., 2020). Nguyen and Tran (2021) propose a backdoor pattern which is a small change in the shape of an object. A perturbation can also be the angle from which a traffic sign is seen (Dong et al., 2022). We conclude with two papers proposing real-world attacks. Sitawarin et al. (2018) use lenticular print, and Nassi et al. (2019) project (slightly altered) traffic signs onto surfaces (Fig. 2(f)).

Table 3

Image classification security (subset). Perturbations applied are global (L_∞), patch-(P) or object-based (O) perturbations on the input image. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

	Reference	Attacker's knowledge			Attacker's capability										E	S
		ω	D	f	X	Y	X_t	δ	targ.	exe.	univ.	transf.	phy.	con.		
Backdoor	Badnets (Gu et al., 2017)	○	○	○	○	○	○	P	✓	✓	✓				○	○
	Wanet (Nguyen and Tran, 2021)	○	○	○	○	○	○	O	✓	✓	✓		✓		○	○
	Yi et al. (2023)	●	○	●	○	○	○	L_∞	✓	✓	✓				○	○
	Patel et al. (2020)	●	○	●	○	●	○	O/P	✓*	✓	✓				○	○
Evasion	Sayles et al. (2021)	○	○	○	●	●	○	L_∞	✓				✓		○	○
	OPAD (Gnanasambandam et al., 2021)	○	○	○	●	●	○	L_∞	✓/				✓		○	○
	ViewFool (Dong et al., 2022)	○	○	○	●	●	○	V				✓	✓		○	○
	Worzyk et al. (2019)	○	○	○	●	●	○	O	✓				✓		○	○
	Yan et al. (2023)	○	○	○	●	●	○	P		✓*	✓		✓		○	○
	Liu et al. (2019)	○	○	○	●	●	○	O	✓/						○	○
	DT-UAPs (Benz et al., 2020)	○	○	○	●	●	○	L_∞ /P	✓	✓	✓				○	○
	AdvCam (Duan et al., 2020)	○	○	○	●	●	○	O	✓/	✓	✓		✓		○	○
	DARTS (Sitawarin et al., 2018)	○/●	○	○/●	●	●	○	O		✓*		✓	✓		○	○
	IAP (Bai et al., 2021)	○/●	○	○/●	●	●	○	P		✓*		✓	✓		○	○
	EAPSO (Jiang et al., 2020)	●	○	○	●	●	○	L_∞							○	○
	Zhong et al. (2022)	●	○	○	●	●	○	P		✓*			✓		○	○
	AdvSL (Yufeng et al., 2023)	●	○	○	●	●	○	O		✓*	✓	✓	✓		○	○
	AdvNB (Hu et al., 2024)	●	○	○	●	●	○	O				✓	✓		○	○
	Huang et al. (2023a)	●	○	○	●	●	○	P							○	○
	AdvLB (Duan et al., 2021)	●	○	○	●	●	○	O					✓		○	○
Poisoning	Yan et al. (2022)	●	○	●	●	●	○	P	✓*	✓*	✓				○	○
	DARTS (Sitawarin et al., 2018)	●	●	●	●	●	○	O	✓	✓*	✓		✓		○	○
	Mobilbye (Nassi et al., 2019)	●	●	●	●	●	○	O	✓/	✓*	✓		✓		○	○
Poisoning	PAPSO (Jiang et al., 2020)	●	○	○	○	●	●	L_∞							○	○

Physical constraints. Many attacks incorporate physical constraints to increase the likelihood of performing well in the problem space. Some of the previous perturbations are, for example, in the shape of laser beams (Duan et al., 2020; Yan et al., 2022), shadows (Zhong et al., 2022), billboards (Patel et al., 2020), or based on rotations (Dong et al., 2022) or small changes in shape (Nguyen and Tran, 2021) and have built-in physical constraints. Other perturbations are executable via LEDs (Sayles et al., 2021) or projectors (Nassi et al., 2019; Gnanasambandam et al., 2021). Otherwise, expectation over transformation (EoT) to incorporate distortions like rotation, scaling, and translation is used (Sitawarin et al., 2018; Zhong et al., 2022; Duan et al., 2020; Bai et al., 2021; Sayles et al., 2021; Worzyk et al., 2019). Additional distortions, like ambient brightness (Sitawarin et al., 2018), random backgrounds (Duan et al., 2020), or addition only (Worzyk et al., 2019) can also be incorporated the loss.

Exposure. Both backdoor and poisoning attacks exhibit low exposure: none was evaluated in practice. Of the evasion attacks, Morgulis et al. (2019) tested DARTS (Sitawarin et al., 2018) on a commercial, unnamed car. Nassi et al. (2019) evaluated their attack on Apollo Mobileye, a commercial system. Furthermore, many attacks printed their perturbations (Bai et al., 2021; Duan et al., 2020; Sayles et al., 2021; Huang et al., 2023a) or implemented them based on shadows (Zhong et al., 2022), lasers (Duan et al., 2020; Yan et al., 2022; Yufeng et al., 2023), LEDs (Sayles et al., 2021), or projectors (Gnanasambandam et al., 2021; Worzyk et al., 2019), and tested them in the problem space or the real world.

Attack severity and qualitative risk. Yufeng et al. (2023) proposed the only attack with the highest severity. In general, both severity and exposure are high, warranting a high overall risk. The number of analyzed models, 17, and data sets, 11, are diverse and thus support this assessment.

Conclusion. Image classification is a mature research area with many publications. Approaches vary concerning attacks, threat models, perturbations, and attack properties. Yet, the field would benefit from more work in poisoning, vehicle-specific privacy assessments, or transferability studies. Despite more than half of the attacks evaluated in the problem space, there are few attacks with the highest severity S.

4.1.2. Action recognition

In action recognition, we predict based on a video sequence what that what activity the person in the video is engaged in (jumping, walking, etc.) (Mordan et al., 2021; Xiong et al., 2022). Action recognition supports the understanding of pedestrians. The task can be implemented as a classification problem on video sequences and is based only on camera inputs, as visible in Table 4.

Attacker's goal. Current attacks on action recognition are limited to integrity attacks at training and test time. During training time, backdoor patterns are correlated with a particular target class (Li et al., 2024; Hammoud et al., 2023; Zhao et al., 2020). At test

Table 4

Action recognition security. Attacker targets apply global (L_∞) or patch (P) perturbations on the image. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

		Attacker's knowledge		Attacker's capability						targ.	exe.	univ.	transf.	E	S
		Reference	ω	D	f	X	Y	X_t	δ						
Backdoor	Li et al. (2024)	●	○	●	○	○	○	L_∞	✓	✓	✓			○	○
	Hammoud et al. (2023)	●	○	●	○	○	○	P	✓	✓*	✓			○	○
	Zhao et al. (2020)	○	○	○	○	●	○	P	✓	✓	✓			○	○
Evasion	Pony et al. (2021)	○	○	○	●	●	○	L_∞		✓	✓/	✓		○	○
	Hwang et al. (2021)	○	○	○	●	●	○	L_∞		✓/	✓			○	○
	Kim et al. (2023)	○	○	○	●	●	○	L_∞			✓	✓		○	○
	MSANet (Wu et al., 2023b)	○	○	○	●	●	○	L_∞						○	○
	Mu et al. (2023)	○	○	○	●	●	○	L_∞						○	○
	Wei et al. (2019b)	○	○	○	●	●	○	L_∞	✓	✓/	✓/	✓		○	○
	Li et al. (2019)	○	○	○	●	●	○	L_∞	✓	✓/	✓			○	○
	V-DSA (Jiang et al., 2023)	●	○	○	●	●	○	L_∞	✓			✓		○	○
	Wei et al. (2020)	●	○	○	●	●	○	L_∞	✓					○	○
	V-BAD (Jiang et al., 2019)	●	○	○	●	●	○	L_∞	✓/					○	○
	VRA (Gupta et al., 2022)	●	○	○/●	●	●	○	L_∞				✓		○	○
	U3D (Xie et al., 2022)	●	○	○/●	●	●	○	L_∞		✓	✓	✓		○	○
	Cao et al. (2023b)	●	○	○/●	●	●	○	L_∞	✓					○	○
	MUA (Inkawhich et al., 2018)	○/●	○	○/●	●	●	○	L_∞				✓		○	○

time, analogous to other classification tasks, untargeted attacks are decreasing overall performance (Hwang et al., 2021; Inkawhich et al., 2018; Gupta et al., 2022; Wei et al., 2020). In contrast, targeted attacks force the model to output an attacker-chosen class or action (Wei et al., 2020; Pony et al., 2021; Gupta et al., 2022; Wei et al., 2019b; Cao et al., 2023b).

Perturbations and physical constraints. One backdoor attack applies a global trigger (Li et al., 2024). The remaining two are based on patch triggers (Hammoud et al., 2023; Zhao et al., 2020). All evasion attacks implement global perturbations altering the entire image (L_∞) (Li et al., 2019; Gupta et al., 2022; Hwang et al., 2021; Inkawhich et al., 2018; Wei et al., 2019b). However, for example Li et al. (2019), Mu et al. (2023) and Wei et al. (2019b) point out that not all video frames must be targeted. There are, however, several honorable mentions. Pony et al. (2021) use a physically possible perturbation, flickering. Mu et al. (2023) base their perturbations on a similarity metric inspired by human perception, Cao et al. (2023b) on style transfer. Jiang et al. (2023) implement L_1 based perturbations consistent over the video, but not constrained to an object or region of the images.

Exposure, attack severity, and qualitative risk. Pony et al. (2021) were the only ones to test their attack in the problem space via RGB light bulbs. Attack severity is neither high nor low, yielding a low-risk assessment, which is supported by a decent number of models (24) and a decent amount of datasets (7).

Conclusion. Only video-based action recognition is studied. Despite a large number of papers, only two attacks and almost only global perturbations are investigated. This is further emphasized in the absence of physical constraints, real-world tests, and high attack severity. More work on poisoning, sponge attacks, model stealing, privacy threats, and real-world evaluations are needed.

4.1.3. Object detection

In object detection, we predict which object is in a camera image (Chan et al., 2022; Thys et al., 2019; Wu et al., 2020a), or in LiDAR 3D point cloud (Xiang et al., 2021; Cao et al., 2019; Tu et al., 2020), or in a combination thereof (Abdelfattah et al., 2021; Hallyburton et al., 2022; Nassi et al., 2020). Whereas LiDAR data is three-dimensional, the former inputs may consist of multidimensional images by several cameras. However, we found only attacks on 2-dimensional image data, e.g. images taken by one camera. In terms of detected objects, for autonomous vehicles, we predict the presence of other traffic participants like other vehicles (V), pedestrians (P), or traffic signs and traffic lights (TF). We focus on AV-related object detection (see Table A.15 in the Appendix) and refer the interested reader to Serban et al. (2020) or more recently, Amirkhani et al. (2023), for a general overview.

Attacker's goal. In object detection, most attacks study integrity at training or test time. In both cases, the attacker's goals are similar. Untargeted damage (Wicker and Kwiatkowska, 2019; Li et al., 2021b; Abdelfattah et al., 2021; Suryanto et al., 2022) decreases the model's performance. Specific goals include targeted misclassification (T) (Gu et al., 2017; Kuang et al., 2021; Xiang et al., 2021; Hallyburton et al., 2022), or changing the class of one object to another, for example, a pedestrian classified as a tree. An object can also be created (C) (Zhu et al., 2023a; Ji et al., 2021; Hallyburton et al., 2022), or an existing object can be hidden (H) (Cai et al., 2020; Cao et al., 2023a, 2021; Luo et al., 2023), e.g., removed from the classifier's output. In addition, object detection is the only field where a confidentiality attack has been proposed, where the attacker steals the victim's model (Liang et al., 2022). Finally, sponge attacks slow the model down, for example, by making it output many more objects than present in the image (Shapira et al., 2023; Liu et al., 2023b).

Perturbations and physical constraints. In contrast to the attacker's goals, perturbations and physical constraints depend on the targeted model and sensor. We provide several examples in Fig. 3.

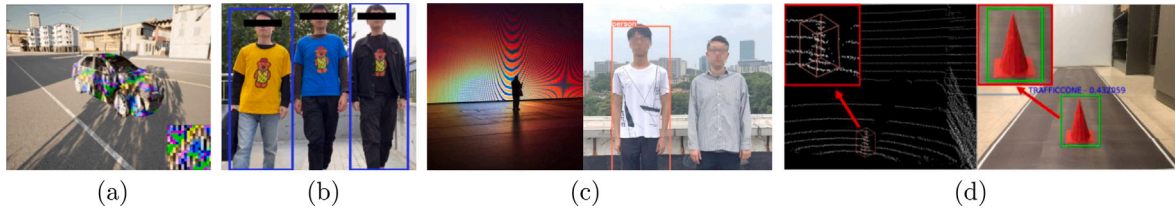


Fig. 3. Specific perturbations proposed for object detection beyond the example from Fig. 2. (a) The vehicle's color is the perturbation (P) (Suryanto et al., 2022). (b) The blue T-shirt with the bear is the backdoor hiding the person (H), similar clothes are benign (Ma et al., 2022). (c) Right: naturally occurring Moiré pattern. Left: exploiting a Moiré pattern caused by striped clothes to hide a person (H) (Wei et al., 2023). (d) The adversarial traffic cone for LiDAR sensors from Cao et al. (2021). Figures taken from referenced papers.

A single paper, by Li et al. (2021b), perturbs the GPS position, using that LiDAR point clouds are functions of the trajectory they are recorded on. While keeping the GPS change smaller than 10 cm, they cause misclassification. On LiDAR data, all perturbations default to being three-dimensional. Perturbations range from changing a subset of the available points (P) (Xiang et al., 2021; Wicker and Kwiatkowska, 2019; Wang et al., 2021a) to forming new objects using points (O) (Cao et al., 2019; Tu et al., 2020). For physical feasibility, EoT (Athalye et al., 2018) can be adapted to point clouds. Perturbations can also be constrained to additions, which are easier to implement physically (Xiang et al., 2021). More practical studies, like the one conducted by Cao et al. (2019) and Jin et al. (2023) investigate the feasibility of spoofing points on LiDAR systems in the real world.

In images, global perturbations cover the entire image (L_∞) (Kuang et al., 2021; Li et al., 2018; Xue et al., 2021). Local perturbations are constrained to a patch, e.g., the shape of the object or a part (square, circle) of the image (P) (Chan et al., 2022; Thys et al., 2019; Jia et al., 2020). To increase the harm of perturbations in practice, EoT (Athalye et al., 2018) or extensions thereof were introduced. Extensions remain within the original color (Cao et al., 2021), transfer style (Zhu et al., 2023a), or limit the number of changed pixels (Man et al., 2020). Due to the number of papers on object recognition, several object-based perturbations have been established. These include 3D car colors for 2D image classification to hide cars (Fig. 3(a)) (Hu et al., 2021; Zhang et al., 2019; Wu et al., 2020b; Suryanto et al., 2022; Wang et al., 2021b), or clothes like T-shirts (Fig. 3(b)) (Ma et al., 2022; Xu et al., 2020) or pullovers (Wu et al., 2020a) to hide people. In particular, clothes can be backdoor triggers (Ma et al., 2022) or fool the model at test time (Xu et al., 2020; Wu et al., 2020a). Another recent perturbation is blur used as test-time perturbation (Ji et al., 2021) or as backdoor trigger (Zhu et al., 2023a). In both cases, the blur is caused by acoustic spoofing on the sensors that stabilize the camera. Another example is a camera's capturing error (or Moiré effect) used as a trigger (Fig. 3(c)) (Wei et al., 2023).

Some attacks on object detection rely on fusing the data of several sensors. Nassi et al. (2020) show that projecting a human or a traffic sign close to the AV, although no other input is received, causes a hard break as it functions as an object creation attack. In contrast, Abdelfattah et al. (2021) and Cao et al. (2021) attack two sensors simultaneously. They first design an adversarial object for LiDAR, to which they add an adversarial texture for vision models (Fig. 3(d)). In contrast, the Frustum attack (Hallyburton et al., 2022) takes advantage that 3D LiDAR signals can be spoofed when the corresponding camera cannot determine the original distance due to depth ambiguity.

Finally, Liang et al. (2022) feed synthetic data to the victim model to steal the model. In a strict sense, they are not altering the data, and do not operate in a physical setting like other attacks.

Exposure. Since many approaches implement physical constraints, exposure is high. Two papers, Eykholt et al. (2018) (tested by Povolny and Trivedi (2020), and Nassi et al. (2020)) evaluate on real Tesla cars. In addition, Cao et al. (2019, 2021, 2023a), Jin et al. (2023) and Ji et al. (2021) test their attacks on production-level components like Apollo mobileye. More than half of the papers perform real-world tests by printing their perturbations and then capturing them (Ma et al., 2022; Xu et al., 2020; Wu et al., 2020a; Zhao et al., 2019; Chen et al., 2018; Wei et al., 2023). LiDAR-based attacks are more difficult to test in practice, yet testbeds can be used (Cao et al., 2023a; Jin et al., 2023).

Attack severity and qualitative risk. There are several attacks with the highest severity, and the exposure of the attacks is also high. The overall risk is thus also high and supported by a large number of models (61) and datasets (22).

Conclusion. A large amount and variety of papers study object detection security, even when restricting the application to AVs. There is a large variety of sensors targeted and threat models. The attacker's goals have been established and elaborated, and most attack properties are well covered. The field is also rather concrete in terms of tests and attack severity. Ideally, more LiDAR-based attacks and street-legal vehicle vulnerabilities would be studied.

4.1.4. Object tracking

Object tracking is similar to object detection, but incorporates the temporal component of following an object over time, as visualized in Fig. 4A. Several paradigms exist to solve this task. Detect-to-track applies object detection on every single image. We discussed an example already in the object detection section (Jia et al., 2020). Furthermore, track-to-detect incorporates information from the previous frame for detection. All papers in this section follow this approach. Recent methods form new paradigms by combining the previous two or relying on attention. Analogous to object detection, object tracking can also track multiple objects or use LiDAR data (Liu et al., 2023a). All approaches except Liu et al. (2023a) process images, as visible in Table 5.

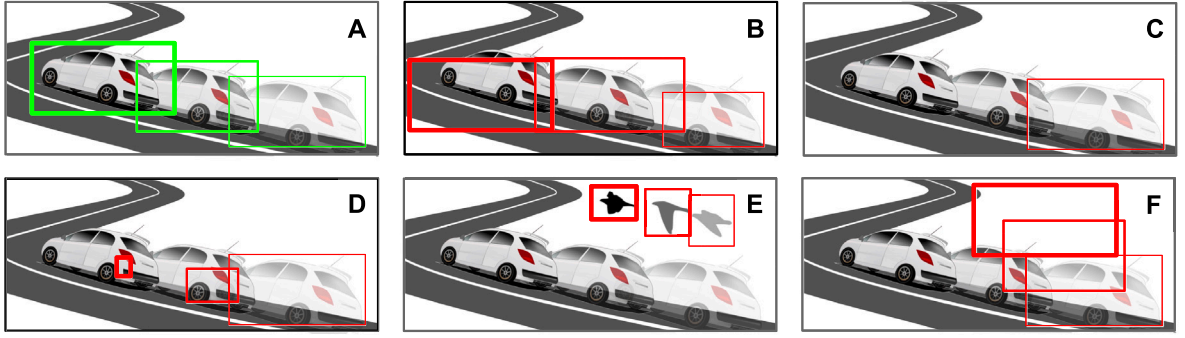


Fig. 4. Attacker's goals in object tracking methods. A: untampered output, the vehicle is correctly tracked over time. B: Untargeted attack decreasing the performance of the tracking accuracy (Li et al., 2022). C: Targeted attack: the tracking algorithm loses the target (H in Table 5) (Chen et al., 2020). D: Targeted attack shrinking the tracked object (S in Table 5) (Yan et al., 2020b). E: Targeted attacks that misguided the model to track another object instead (here a bird, O in Table 5) (Yan et al., 2020a). F: Targeted attack that makes the algorithm follow a specified trajectory instead (T in Table 5) (Liang et al., 2020).

Table 5

Object tracking security. Perturbations are global (L_∞) or patch-based (P). Various targeted attacks have been suggested: hiding or removing an object (H), shrinking the object until it disappears (S), introducing a new object (O) or enforce a predetermined trajectory (T). Threat model, attack properties and exposure E and severity S as in Section 3.

Object tracking			Attacker's knowledge			Attacker's capability												E	S
Sensor		Reference	ω	D	f	X	Y	X_t	δ	targ.	exe.	univ.	transf.	phy.	con.				
Backdoor	Camera	Badtrack (Huang et al., 2023b)	●	○	●	○	○	○	P		✓	✓						○	●
	Camera	FSBA (Li et al., 2022)	○	○	○	○	○	○	P		✓	✓						○	●
	Camera	TAT (Cheng et al., 2023)	○	○	○	○	○	○	P	O	✓	✓		✓				○	●
	Camera	Yan et al. (2020b)	○	○	○	●	●	○	L_∞	S			✓					○	●
Evasion	Camera	EUSA (Liu et al., 2022)	○	○	○	●	●	○	L_∞	/T	✓	✓						○	●
	Camera	PAT (Wiyatno and Xu, 2019)	○	○	○	●	●	○	P	/T	✓*			✓				○	●
	Camera	Chen et al. (2020)	○	○	○	●	●	○	L_∞	H	✓	✓						○	●
	Camera	Yan et al. (2020a)	○	○	○	●	●	○	L_∞	O								○	●
	Camera	SPARK (Guo et al., 2020)	●	○	○	●	●	○	P	/T			✓					○	●
	Camera	FAN (Liang et al., 2020)	○/●	○	○/●	●	●	○	L_∞	/T			✓					○	●
	Camera	Pang et al. (2023)	○/●	○	○/●	●	●	○	L_∞				✓					○	●
	LiDAR	TAN (Liu et al., 2023a)	○/●	○	○/●	●	●	○	L_∞				✓					○	●

Attacker's goal. We visualize the attacker's goals in Fig. 4. Attacks harm only integrity, at training and test time. The training time backdoor attack is untargeted and decreases the accuracy as much as possible (Fig. 4B) (Li et al., 2022; Liu et al., 2023a) or to follow an inserted object (O) (Cheng et al., 2023). There are also untargeted test-time attacks (Wiyatno and Xu, 2019; Liang et al., 2020; Liu et al., 2022). However, more sophisticated evasion goals have also been proposed. Chen et al. (2020) suggest hiding or removing an object (H, Fig. 4C), Yan et al. (2020b) shrink the object until it disappears (S, Fig. 4D). Yan et al. (2020a) aim to track another object than the intended target (O, Fig. 4E). Finally, several attacks force the model to follow a predetermined trajectory (T, Fig. 4F) despite the object's true path (Wiyatno and Xu, 2019; Liang et al., 2020; Guo et al., 2020; Liu et al., 2022).

Perturbations and physical constraints. The backdoor attacks rely on patches (P) (Li et al., 2022; Cheng et al., 2023; Huang et al., 2023b). In evasion, global perturbations (L_∞) (Chen et al., 2020; Liang et al., 2020; Liu et al., 2022; Chen et al., 2020; Yan et al., 2020b) or patch-based perturbations (P) (Wiyatno and Xu, 2019; Guo et al., 2020) are used. No attack considers physical constraints, except PAT (Wiyatno and Xu, 2019) and TAT (Cheng et al., 2023), which rely on variants of EoT. In addition, the PAT attack is camouflaged as a poster.

Exposure, Attack severity, and qualitative risk. The latter two attacks are also tested in the problem space. Wiyatno and Xu (2019) display their poster on a screen for validation, Cheng et al. (2023) recorded a video. In contrast to this overall low exposure, attack severity seems neither high nor low, warranting a low-to-medium overall risk. A decent number of models (11) and datasets (9) support this assessment for camera-based approaches. In the case of LiDAR, three models are tested on one dataset.

Conclusion. While there is a decent amount of articles, the diversity of attacks, studied sensors, and perturbations could be better. Attacks within one attack class do not vary much: evasion attacks introduce mainly L_∞ , backdoors patch-based perturbations. Instead, the attackers' goals and attack properties are somewhat diverse, except for the low number of physical constraints and practical evaluations. More work is also needed to cover poisoning attacks, model stealing, privacy, and sponge attacks to develop a deep understanding of object tracking's vulnerability.

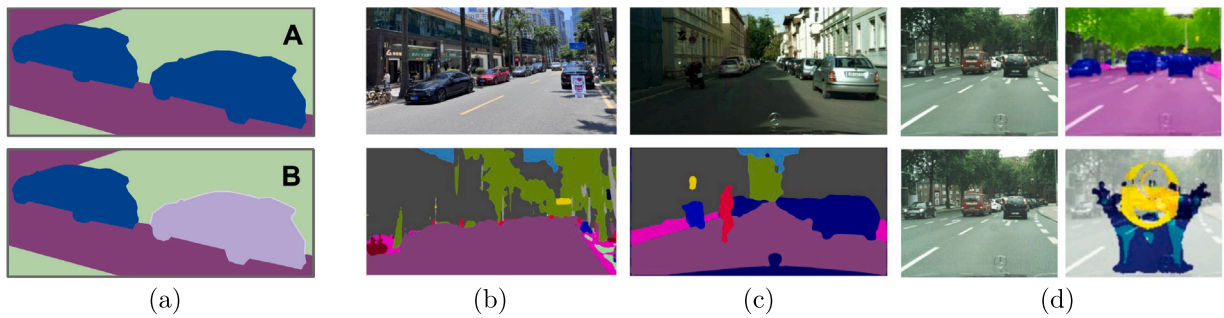


Fig. 5. Semantic and instance segmentation security. (a) difference between semantic segmentation (A) and instance segmentation (B): while the former only outputs the class car (dark blue), the latter distinguishes instances (blue car, violet car). (b) The trigger (Hello Kitty sign on the left) leads to the vehicles being classified as roads. Approach and visualization by [Lan et al. \(2024\)](#). (c) The attack inserts a non-existing pedestrian (red) in the scene's segmentation above. Illustration and attack by [Chen et al. \(2022\)](#). (d) top: left original input, right benign segmentation output. bottom: model outputs a segmentation chosen by the attacker. Attack and images by [Cisse et al. \(2017\)](#).

4.1.5. Image segmentation

We now start discussing pixelwise prediction tasks. The first is image segmentation, of which there are two varieties of image segmentation: semantic segmentation ([Nesti et al., 2022](#); [Arnab et al., 2018](#); [Fischer et al., 2017](#); [Metzen et al., 2017](#)) assigns a label like “car” or “pedestrian” to each pixel. Instance segmentation ([Mao et al., 2023](#); [Li et al., 2018](#)) focuses on objects and distinguishes instances (e.g., “car 1”, “car 2”, etc.), as visualized in [Fig. 5\(a\)](#). All works in the following section are based on camera input, except [Mahima et al. \(2023\)](#), who work on LiDAR data. We summarize all papers in [Table 6](#).

Attacker's goal. We visualize examples for the attacker's goal in [Fig. 5\(b\)–\(d\)](#) using examples from the papers below. There are three backdoor attacks, two on instance, and one on semantic segmentation. All cause the model to output a predetermined segmentation class, e.g., segment a person as a palm tree or as road ([Li et al., 2021a](#); [Mao et al., 2023](#); [Lan et al., 2024](#)). Integrity at test time, or evasion, has been studied exhaustively in image segmentation. Many works attempt to increase the loss at test time ([Arnab et al., 2018](#); [Shen et al., 2019](#); [Kang et al., 2020](#); [Gu et al., 2022](#); [Nesti et al., 2022](#); [Rossolini et al., 2023](#)), or in other words, decrease the quality of the model's segmentation output. As a slightly more ambitious goal, [Fischer et al. \(2017\)](#) and [Nakka and Salzmann \(2020\)](#) attempt to change the model's output to the neighboring class, the background class ([Rony et al., 2023](#)), or another class ([Fig. 5\(b\)](#)) ([Croce et al., 2023](#)). More ambitious attacks change the segmentation to the dataset's average ([Rony et al., 2023](#)), make the model hallucinate an object ([Fig. 5\(c\)](#)) ([Chen et al., 2022](#)), or make the model output a desired segmentation ([Fig. 5\(d\)](#)) ([Metzen et al., 2017](#); [Xu et al., 2020](#)).

Finally, three works study privacy, or membership inference, on semantic segmentation ([He et al., 2020](#); [Shafran et al., 2021](#); [Zhang et al., 2022b](#)). They vary the knowledge of the attacker: while [He et al. \(2020\)](#) and [Shafran et al. \(2021\)](#) access the full confidence scores, [Zhang et al. \(2022b\)](#) show the model leaks membership of its data even under hard labels or only class output.

Perturbations. The backdoor trigger is an object (O), e.g., a wall, or a black line ([Li et al., 2021a](#); [Lan et al., 2024](#)), or a small patch (P) ([Li et al., 2021a](#); [Lan et al., 2024](#)). In evasion or test time integrity, the majority of perturbations are global (L_∞), covering the entire image ([Cisse et al., 2017](#); [Metzen et al., 2017](#); [Xie et al., 2017](#); [Li et al., 2018](#); [Arnab et al., 2018](#); [Shen et al., 2019](#); [Xu et al., 2020](#); [Gu et al., 2021, 2022](#); [Chen et al., 2022](#); [Rony et al., 2023](#); [Croce et al., 2023](#)). In contrast, [Kang et al. \(2020\)](#) and [Fischer et al. \(2017\)](#) constrain the area of the perturbations to objects (O). [Nakka and Salzmann \(2020\)](#), [Nesti et al. \(2022\)](#), and [Rossolini et al. \(2023\)](#) propose patch-based attacks (P), the latter two in the shape of billboards.

Physical constraints and exposure. Very few attacks incorporate physical constraints, where perturbations are either objects like walls ([Li et al., 2021a](#)) or billboards ([Kang et al., 2020](#); [Rossolini et al., 2023](#)) or are constrained to the shapes of cars and pedestrians ([Fischer et al., 2017](#); [Kang et al., 2020](#)). There are no evaluations in the problem space except [Nesti et al. \(2022\)](#) and [Rossolini et al. \(2023\)](#). Both printed their patch and successfully evaluated the pictures of the attack.

Attack severity and qualitative risk. There are papers with high severity, notably [Mirsky \(2023\)](#), but also other evasion ([Nesti et al., 2022](#); [Nakka and Salzmann, 2020](#)) or backdoor attacks ([Mao et al., 2023](#); [Li et al., 2021a](#); [Lan et al., 2024](#)). Due to few tests, the overall risk is medium but seems reliable as many different models, in total 34, and twelve datasets were tested, at least in the case of camera sensors. For LiDAR, [Mahima et al. \(2023\)](#) test five models on one dataset.

Conclusion. Despite the large number of works on semantic segmentation, the variety in studied attacks and applied perturbations is rather low. More work in terms of LiDAR sensors, instance segmentation and overall in model stealing, training time, and sponge attacks are needed. In contrast, studies about the attacker's goals, transferability, and universal attacks exist. Yet, more work is needed to assess practical image and instance segmentation security.

4.1.6. Depth estimation

In depth estimation, we regress the distance of each pixel to the camera. While approaches based on multiple cameras exist, all papers listed in [Table 7](#) use a single camera's footage.

Attacker's goal. All attacks harm the model's integrity at test time and are evasion attacks. Untargeted attacks cause wrong outputs ([Zhang et al., 2020b](#)). More ambitious goals encompass setting a specific depth for an object ([Zhang et al., 2020b](#)), increasing

Table 6

Image segmentation security. Attacker targets instance (inst.) or semantic (sem.) segmentation, and perturbations on input images are global (L_∞), patch-(P), or object-shaped (P). Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

Appr.	Attacker's knowledge				Attacker's capability							targ.	exe.	univ.	transf.	phy.	con.	E	S
	Reference	ω	D	f	X	Y	X_t	δ											
Backdoor	inst.	OFBA (Mao et al., 2023)	●	○	●	○	○	○	P	✓	✓	✓						○	●
	sem.	Li et al. (2021a)	●	○	●	○	○	○	O	✓	✓	✓/			✓			○	●
	sem.	IBA (Lan et al., 2024)	●	○	●	○	○	○	O/P	✓	✓	✓						○	●
Evasion	inst.	RAP (Li et al., 2018)	○/●	○	○/●	●	●	○	L_∞	✓								○	○
	sem.	Nesti et al. (2022)	○	○	○	●	●	○	O/P		✓*	✓			✓			○	●
	sem.	Nakka and Salzmann (2020)	○	○	○	●	●	○	P	✓	✓/	✓/						○	●
	sem.	Xu et al. (2020) and Rony et al. (2023)	○	○	○	●	●	○	L_∞	✓/								○	○
	sem.	SegPGD (Gu et al., 2022)	○	○	○	●	●	○	L_∞									○	○
	sem.	Metzen et al. (2017)	○	○	○	●	●	○	L_∞	✓		✓	✓					○	○
	sem.	Fischer et al. (2017)	○	○	○	●	●	○	L_∞ /O	✓						✓		○	○
	sem.	Croce et al. (2023)	○	○	○	●	●	○	L_∞	✓								○	○
	sem.	Rossolini et al. (2023)	○	○	○	●	●	○	O			✓*				✓		○	○
	sem.	Houdini (Cisse et al., 2017)	○	○	○	●	●	○	L_∞	✓/								○	○
	sem.	AdvSPaDE (Shen et al., 2019)	●	○	○	●	●	○	L_∞									○	○
	sem.	Arnab et al. (2018) and (Gu et al., 2021)	○/●	○	○/●	●	●	○	L_∞						✓			○	○
	sem.	Kang et al. (2020)	○/●	○	○/●	●	●	○	L_∞ /O						✓	✓		○	○
	sem.	Chen et al. (2022)	○/●	○	○/●	●	●	○	L_∞	✓					✓			○	○
	sem.	Mirsky (2023)	○/●	○	○/●	●	●	○	P	✓	✓*	✓	✓	✓	✓			○	○
sem.	DAG (Xie et al., 2017; Mahima et al., 2023)	○/●	○	○/●	●	●	○	L_∞						✓			○	○	
Privacy	sem.	He et al. (2020), Zhang et al. (2022b) and Shafran et al. (2021)	●	○	○	●	○	○	-	-	-	-	-	-	-	-		○	○

Table 7

Depth estimation security. Attacks target position- or camera-based approaches, and apply global (L_∞), object shaped (O), or patch-based (P) perturbations on the image. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

		Attacker's knowledge		Attacker's capability													
		Reference	ω	D	f	X	Y	X_l	δ	targ.	exe.	univ.	transf.	phy.	con.	E	S
Evasion	Cheng et al. (2022)	○	○	○	●	●	○	P			✓*	✓		✓		○	○
	Yamanaka et al. (2020)	○	○	○	●	●	○	P	✓	✓	✓			✓		○	○
	APARATE (Guesmi et al., 2023)	○	○	○	●	●	○	P	✓	✓*	✓			✓		○	○
	Zhang et al. (2020b)	○	○	○	●	●	○	L_∞	✓/	✓*	✓/					○	○
	Daimo and Ono (2023)	●	○	○	●	●	○	O	✓							○	○
	Wong et al. (2020)	○/●	○	○/●	●	●	○	L_∞ /P	✓				✓			○	○

the object's depth (Cheng et al., 2022), choosing an extreme low or high depth (Yamanaka et al., 2020), flipping scenes (Wong et al., 2020), matching another scene's the depth (Wong et al., 2020), or hiding objects (Guesmi et al., 2023; Daimo and Ono, 2023).

Perturbations. There are global, image-wide perturbations (L_∞) (Wong et al., 2020; Zhang et al., 2020b). In addition, small, patch-based perturbations (P) (Yamanaka et al., 2020; Cheng et al., 2022; Wong et al., 2020) and object-shaped perturbations (O) (Daimo and Ono, 2023) have been proposed.

Physical constraints. Only three attacks (Yamanaka et al., 2020; Zhang et al., 2020b; Guesmi et al., 2023) implement physical constraints related to EoT. In addition, Yamanaka et al. (2020) consider brightness and printability and Cheng et al. (2022) match the style of traffic barriers.

Exposure, Attack severity and qualitative risk. Yamanaka et al. (2020) print and photograph their attack, Daimo and Ono (2023) project their perturbations on the target object. In contrast to this low exposure, attack severity is rather high, yielding a medium risk assessment. While several (8) models are tested, only two datasets are used. For a useful risk assessment, more work is needed.

Conclusion. Few works study the security of depth estimation, and all study evasion on only one dataset. There is however a large variety in the attack goals and studied properties. While there is some consideration for the physical feasibility of the attacks and transferability, more tests would be beneficial. Finally, more work is needed to understand training time vulnerabilities like backdoors and poisoning, privacy risks, model stealing, and sponge attacks.

Table 8

Optical flow security. Perturbations are either global (L_∞), patch-(P) or object-shaped (O). Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

		Attacker's knowledge			Attacker's capability											
		Reference	ω	D	f	X	Y	X_t	δ	targ.	exe.	univ.	transf.	phy. con.	E	S
Evasion	Agnihotri and Keuper (2023)		○	○	○	●	●	○	L_∞	✓					○	○
	Schmalfuss et al. (2023)		○	○	○	●	●	○	L_∞		✓	✓		✓	○	○
	Wortman (2021)		○	○	○	●	●	○	P		✓	✓		✓	○	○
	Schrodi et al. (2022)		●	○	●	●	●	○	P		✓	✓			○	○
	Ranjan et al. (2019)		○/●	○	○/●	●	●	○	P	✓	✓	✓		✓	○	○
	Koren et al. (2022)		○/●	○	○/●	●	●	○	L_∞/O				✓		○	○
	MUA (Inkawhich et al., 2018)		○/●	○	○/●	●	●	○	L_∞				✓		○	○
	Schmalfuss et al. (2022)		○/●	○	○/●	●	●	○	L_∞	✓	✓	✓	✓		○	○

Table 9

Camera pose estimation security. Perturbations are either global (L_∞) or patches (P) on the input image. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

		Attacker's knowledge			Attacker's capability											
		Reference	ω	D	f	X	Y	X_t	δ	targ.	exe.	univ.	transf.	phy. con.	E	S
Evasion	Xie et al. (2023)		○	○	○	●	●	○	L_∞	✓	✓	✓			○	○
	Chawla et al. (2022)		○	○	○	●	●	○	L_∞	✓/			✓		○	○
	Nemcovsky et al. (2022)		○	○	○	●	●	○	P		✓	✓		✓	○	○

4.1.7. Optical flow

Optical flow prediction determines for each pixel in an image sequence velocity and direction of movement (Inkawhich et al., 2018; Ranjan et al., 2019; Schrodi et al., 2022). This supports, for example, computing a possible position or time until collision (Agnihotri and Keuper, 2023). All existing works study camera-based approaches, as summarized in Table 8.

Attacker's goal and perturbations. Only attacks targeting integrity, more specifically evasion attacks have been studied. Untargeted attacks decrease performance (Inkawhich et al., 2018; Wortman, 2021; Koren et al., 2022; Schrodi et al., 2022); targeted attacks invert the computed flow (Ranjan et al., 2019), target non-existing flow (Schmalfuss et al., 2022) or a specific flow output (Agnihotri and Keuper, 2023). Perturbations to reach these goals are image-wide, global perturbations (L_∞) (Inkawhich et al., 2018; Schmalfuss et al., 2022; Agnihotri and Keuper, 2023; Schmalfuss et al., 2023), patches (P) (Ranjan et al., 2019; Wortman, 2021; Schrodi et al., 2022), or can also be constrained to fit an object (O) like a car (Koren et al., 2022).

Physical constraints. Most approaches do not consider physical constraints. Exceptions are Ranjan et al. (2019), who use EoT, and Schmalfuss et al. (2023) with weather-based perturbations. Wortman (2021) suggest a transparent patch to ensure the perturbation is applicable as a sticker to any surface.

Exposure, attack severity, and qualitative risk. Only one paper evaluates in the problem space: Ranjan et al. (2019) print their perturbations. Attack severity covers a medium to high range, warranting a medium qualitative risk assessment. While the number of models is high (12), only four datasets support these results, questioning the reliability of our assessment.

Conclusion. Although a decent amount of papers provide some variance within the attacker's goal and implemented perturbations, there is no variance in attacks and sensors and few real-world evaluations. More work is needed to understand training time, confidentiality, and sponge vulnerabilities, and real-world risks.

4.1.8. Camera pose estimation

Pose estimation regresses, from visual input, the camera position in terms of rotation and translation (Nemcovsky et al., 2022; Chawla et al., 2022; Xie et al., 2023). All attacks found are based on camera images and are summarized in Table 9.

Attacker's goal and perturbations. Only evasion attacks have been studied. Untargeted attacks make the model deviate from its real position (Nemcovsky et al., 2022; Chawla et al., 2022). Targeted attacks, for example by Chawla et al. (2022) or Xie et al. (2023), make the mode invert the computed movement, or invert the perceived motion or pose. Perturbations to reach these goals are image-wide, global perturbations (L_∞) (Chawla et al., 2022; Xie et al., 2023), or patches (P) (Nemcovsky et al., 2022).

Physical constraints. To improve the practical applicability of the patches, analogous to many image-based attacks, Nemcovsky et al. (2022) use EoT.

Exposure, attack severity and qualitative risk. The same authors (Nemcovsky et al., 2022) also evaluate in the problem space using a drone. Overall, attack severity is medium, warranting a medium qualitative risk assessment. While the number of models is medium (5), only two datasets support these results, questioning the reliability of our result.

Conclusion. Few works investigate camera pose estimation, with a complete absence of backdoor, poisoning, model stealing, privacy, and sponge attacks. There is little diversity of threat models and datasets, and more real-world settings need to be studied. This is also reflected in the studied perturbations: while there are patches, no object-shaped perturbations are studied.

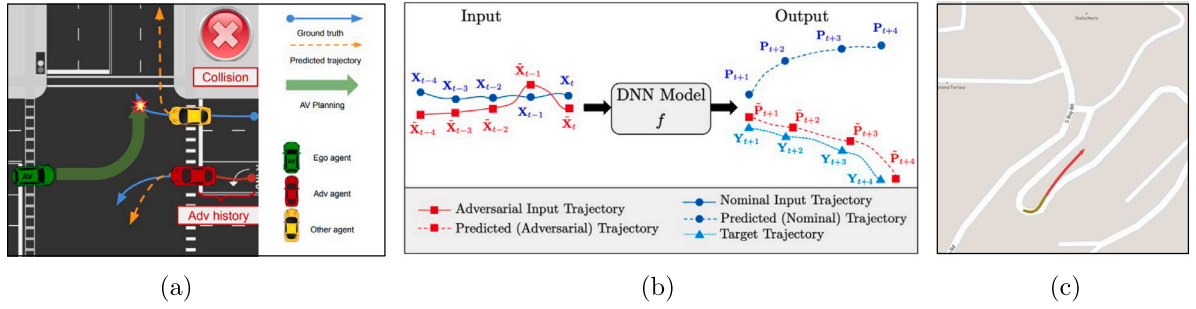


Fig. 6. Trajectory prediction security. (a) Illustration about trajectory prediction security by Cao et al. (2022). The AV (green car) will cause an accident due to a wrong prediction (yellow dashed arrow) because of the adversarial agents' (red car) adversarial history (red arrow). (b) Tan et al. (2023) also visualize the adversarial attack, depicting the input past coordinates into the DNN model, leading to a series of predicted coordinates for the future trajectories. (c) Bahari et al. (2022) do not perturb trajectories but instead find real-world scenes where the agent fails.

4.2. Prediction

Prediction is the pillar following perception, fewer articles cover this area. In contrast to perception, where the AV aims to understand the world around it, prediction aims to understand how the world around it will change *in the future*. In this pillar, we review trajectory prediction. Pedestrian intention prediction (Razali et al., 2021; Guo et al., 2022) (e.g., recognizing when a pedestrian aims to cross the street) has not been evaluated in terms of security yet.

4.2.1. Trajectory prediction

Trajectory prediction regresses the future trajectories of vehicles (Bahari et al., 2021; Zhang et al., 2023) and vulnerable traffic participants such as pedestrians (Razali et al., 2021; Kothari et al., 2021c,a; Saadatnejad et al., 2022). Attacks follow two main approaches. Most work on coordinates (Fig. 6(a) and (b)) (Zhang et al., 2022a; Cao et al., 2022; Tan et al., 2023; Saadatnejad et al., 2022; Zheng et al., 2023; Pourkeshavarz et al., 2023; Messaoud et al., 2023), where a graph or a rasterized image represents the scene and contains information about car positions, lanes, or sidewalks. The second approach directly takes the scene or map representation as input (Fig. 6(c)) (Bahari et al., 2022). We give an overview in Table 10.

Attacker's goal. There are only attacks targeting integrity. At training time, targeted backdoors cause an unreasonable emergency break in AVs (Pourkeshavarz et al., 2023; Messaoud et al., 2023). Untargeted evasion attacks make the AV leave the lane or road (Zhang et al., 2022a; Cao et al., 2022; Bahari et al., 2022; Zheng et al., 2023) or cause collisions (Saadatnejad et al., 2022). Targeted evasion attacks (Tan et al., 2023) force the AV to follow an attacker-chosen trajectory.

Perturbations. To achieve these goals, several perturbations have been explored. Most perturb the trajectories (T) (Pourkeshavarz et al., 2023; Messaoud et al., 2023; Saadatnejad et al., 2022) or coordinates of another, attacker-controlled vehicle within the scene (O) (Pourkeshavarz et al., 2023; Messaoud et al., 2023; Cao et al., 2022) or the victim's car itself (S) (Tan et al., 2023; Saadatnejad et al., 2022). Alternatively, the visual input to the algorithm can be perturbed (I) (Bahari et al., 2022).

Physical constraints. Visual changes are usually applied in the digital domain (Zheng et al., 2023). However, Bahari et al. (2022) generate an entire scene matching an existing scene where the algorithm will fail. For coordinate-based attacks, the trajectory must be drivable. Recent works thus incorporate drivability constraints, either by adapting the loss (Cao et al., 2022; Pourkeshavarz et al., 2023), manually checking the trajectories (Zhang et al., 2022a; Pourkeshavarz et al., 2023; Messaoud et al., 2023), or by constraining to trajectories to be already present in the dataset (Tan et al., 2023; Messaoud et al., 2023).

Exposure, attack severity, and qualitative risk. No attack evaluates in the problem space, with the notable exception of Cao et al. (2022), who tested their work within an AV pipeline. Attack severity is overall medium, warranting a low-to-medium risk assessment. While a reasonable number of eighteen models are tested, the number of datasets is with eight medium.

Conclusion. With only eight papers and eight studied datasets, the security assessment in trajectory prediction is limited. More work is needed to assess vulnerability towards poisoning, sloth attacks, model stealing, and privacy attacks. Most existing attacks target coordinate-based approaches for vehicles, and few consider scene-based methods. In addition, none of the attacks have been evaluated in the problem space. Yet, physical constraints are considered, yielding rather practical threat models.

4.3. Planning

After the previous pillars, the planning component follows, which only encompasses one individual task. Planning determines a path for an AV, given a start and an endpoint. Global planning searches a path between two discrete points on a map (Ma et al., 2019; Yang et al., 2020a; Zhang et al., 2020a; Mo et al., 2022). In contrast, local planning avoids obstacles (Vemprala and Kapoor, 2021). In addition, planning input is frequently an image of a map (Lin et al., 2017; Yang et al., 2020a; Mo et al., 2022) or discrete

Table 10

Trajectory prediction security. Attacks target coordinates (coord.) or scenes, and apply perturbations to the input image (I) or the trajectory (T). In the latter case, we distinguish perturbing the trajectory of another vehicle (O) or the victim's vehicle (S). Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

Trajectory pred.	Attacker's knowledge																Attacker's capability					
	Appr.	Reference	ω	D	f	X	Y	X_t	δ	targ.	exe.	univ.	transf.	phy. con.	E	S						
Backdoor	coord.	Pourkeshavarz et al. (2023)	●	○	○	○	○	○	T	O	✓	✓		✓	🟡	🟡						
	coord.	Pourkeshavarz et al. (2023)	○/●	○	○/●	○	○	○	T	O	✓	✓		✓	🟡	🟡						
	coord.	Messaoud et al. (2023)	●	○	●	○	○	○	T	O	✓	✓/	✓	✓	🟡	🟡						
Evasion	coord.	TA4TP (Tan et al., 2023)	○	○	○	●	●	○	T	S	✓	✓		✓	🟡	🟡						
	coord.	Saadatnejad et al. (2022)	○	●	○	●	●	●	T	S		✓			🟡	🟡						
	coord.	Zhang et al. (2022a)	○/●	○	○	●	●	○	T	O		✓/		✓	🟡	🟡						
	coord.	Zheng et al. (2023)	○/●	○	○	●	●	○	I			✓/			🟡	🟡						
	coord.	AdvDO (Cao et al., 2022)	○/●	○	○/●	●	●	○	T	O		✓/		✓	🟡	🟡						
	Scene	Bahari et al. (2022)	●	●	○	●	●	●	I			✓	✓	✓	🟡	🟡						

and based on states (Ma et al., 2019; Zhang et al., 2020a; Vemprala and Kapoor, 2021). An additional distinction is the amount of agents considered in the paper. This number ranges from 1 (Ma et al., 2019; Yang et al., 2020a) over 2 (Mohammadi et al., 2023) to several (Lin et al., 2017; Mo et al., 2022). We give an overview of all attacks on planning in Table 11.

Attacker's goal and perturbations. Two integrity attacks at training time or backdoors target global vision-based planners. Both make the victim learn two policies, where a trigger activates the bad policy (Yang et al., 2020a; Mo et al., 2022). A trigger can be a colored pixel in the map (P) (Yang et al., 2020a); or a global perturbation on the image input (L_∞) (Mo et al., 2022).

Four evasion attacks have been studied. One on a global, vision-based, two on a local, state-based local planner, and one on a local planner based on both states and visual input. Global perturbations (L_∞) decrease performance (Ren et al., 2023) or cause the agent to perform a desired sequence of actions (Lin et al., 2017), for example, outputting a specific path. On the other hand, Rempe et al. (2022) and Hanselmann et al. (2022) find trajectories of other agents that increase the likelihood or directly cause collisions.

Several training time attacks target state-based planners. Four of these cause the global planner to learn a target policy (Ma et al., 2019; Mohammadi et al., 2023) or a policy partially defined by the attacker (Zhang et al., 2020a; Xu et al., 2021). To this end, the adversary applies sparse (L_1) (Ma et al., 2019) or global (L_∞) (Zhang et al., 2020a) perturbations on the reward R . Alternatively, a peer agent (F_o) is tampered with Mohammadi et al. (2023). Xu et al. (2021) achieve the same goal by tampering with the environment's hyperparameters ($\mathcal{E}$). Vemprala and Kapoor (2021) attempt to make the agent crash with obstacles or the adversarial agent. They position the adversarial agent to make the planner fail or to increase the needed time to compute a solution, or a sponge attack. In contrast to backdoors, none of these attacks inserts a trigger pattern during test time.

Finally, one work studies confidentiality in a global planning setting with a single agent. This work shows that a policy may leak the maps it was trained on. As this work submits samples and classifies members based on confidence, we do not discuss its practicality in the following.

Physical constraints, exposure, attack severity, and qualitative risk. Few attacks constrain perturbations, none evaluate in the problem space or real world. Analogously, attack severity is in general low, yielding a low risk. As an honorable mention, Ren et al. (2023) draw the noise distribution for their perturbation from real-world noise as observed in sensors. In addition, Rempe et al. (2022) and Hanselmann et al. (2022) use a bicycle model to obtain physically possible trajectories. A decent number of models (13) and a borderline amount of datasets (7) support this analysis.

Conclusion. Most papers focus on global planning and state based approaches. In contrast to other tasks, there are few evasion and privacy attacks but many training time attacks, of which one is also a sponge attack. There is variety in the number of agents, however much less overall variance within attacks. Furthermore, we find a complete absence of transferability studies (probably related to the lack of evasion attacks, where transferability is commonly studied). Finally, there are no real-world evaluations and no considerations of physical constraints.

4.4. Control or lane keeping

Many control tasks can be directly tackled by deterministic systems, however, one example that has been studied in terms of AI security is lanekeeping.

Lane keeping is a regression task, where the goal is to stay in the middle of the lane. Given, for example, a video of the road (Zhou et al., 2020; Sato et al., 2021; Jing et al., 2021; Gong et al., 2022; Han et al., 2022) or the distance to the lane and the lane angle (Farivar et al., 2021), we predict the continuous steering angle that keeps the AV centered in the lane. Both ML (Zhou et al., 2020; Sato et al., 2021; Jing et al., 2021; Han et al., 2022; Song et al., 2023; Guo et al., 2023) and RL (Boloor et al., 2019; Sun et al., 2020; Farivar et al., 2021; Gong et al., 2022; von Stein et al., 2023) approaches can solve this task, as visible in Table 12.

Table 11

Planning security. The attacker targets global (route finding) or local (collision avoidance) planning based on camera inputs or states. The perturbations are global (L_∞), sparse (L_1) or patch-based (P) on the input image, reward $\mathcal{R}$, environment $\mathcal{E}$, or a peer agent $\mathcal{F}_o$. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

Planning	Task	# Agents	Appr.	Reference	Attacker's knowledge			Attacker's capability				targ.	exe.	univ.	phy.	con.	S
					ω	D	f	X	Y	X_t	δ						
Backdoor	Global	≥ 1	Camera	DAP (Mo et al., 2022)	○	○	○	○	○	○	L_∞	✓		✓			○
	Global	1	Camera	Yang et al. (2020a)	●	○	●	○	○	○	P	✓	✓/	✓			●
Poisoning	Global	1	State	Zhang et al. (2020a)	○	○	○	○	○	●	L_∞	$\mathcal{R}$					○
	Global	1	State	Ma et al. (2019)	●	○	●	○	○	●	L_1	$\mathcal{R}$	✓				○
	Global	1	State	TEPA (Xu et al., 2021)	○/●	○	○	○	●	●		$\mathcal{E}$	✓				○
	Global	2	State	Mohammadi et al. (2023)	●	○	●	○	●	●	L_∞	$\mathcal{F}_o$	✓		✓		●
	Local	1	State	Vemprala and Kapoor (2021)	●	○	○	●	●	○		$\mathcal{E}$		✓			●
Evasion	Global	≥ 1	Camera	Lin et al. (2017)	○	○	○	●	●	○	L_∞	✓					○
	Local	> 1	State	Ren et al. (2023)	○	○	○	●	●	○	L_∞			✓		✓	●
	Local	≥ 1	Both	KING (Hanselmann et al., 2022)	○	○	○	●	●	○		$\mathcal{E}$		✓		✓	●
	Local	≥ 1	State	STRIVE (Rempe et al., 2022)	●	○	○	●	●	○		$\mathcal{E}$	✓	✓		✓	●
Privacy	Global	1	Both	Pan et al. (2019)	●	○	○	●	●	○	-	-	-	-	-	-	○

Table 12

Lane keeping security. Attacker targets ML or RL; camera or motion sensors, and applies global (L_∞) patch-(P) or object-based (O) perturbations on the input image, observation o , environment $\mathcal{E}$ or reward $\mathcal{R}$. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

Lane keeping			Attacker's knowledge					Attacker's capability					targ. exe. univ. phy. con. E S				
			Reference	ω	D	f	X	Y/R	X_t	δ							
Backdoor	ML	Camera	Han et al. (2022)	●	○	●	○	●	○	O		✓*	✓	✓			
	RL	Camera	BAFFLE (Gong et al., 2022)	●	○	●	○	○	○	P	s/R	✓	✓	✓			
Evasion	ML	Camera	DRP Attack (Sato et al., 2021)	○	○	○	●	●	○	P		✓*	✓	✓			
	ML	Camera	Deepbillboard (Zhou et al., 2020)	○	○	○	●	●	○	O		✓	✓*		✓		
	ML	Camera	Tian et al. (2018)	○	○	○	●	●	○	L_∞					✓		
	ML	Camera	Chernikova et al. (2019)	○	○	○	●	●	○	L_∞		✓					
	ML	Camera	von Stein et al. (2023)	○	○	○	●	●	○	O		✓	✓*		✓		
	ML	Camera	Jing et al. (2021)	●	○	●	●	●	○	P							
	ML	Camera	Song et al. (2023)	●	○	●	●	●	○	O			✓		✓		
	ML	Fusion	Guo et al. (2023)	○	○	○	●	●	○	P		✓	✓*		✓		
	RL	Camera	Critical point attack (Sun et al., 2020)	○	○	○	●	●	○	L_∞	o		✓				
	RL	Camera	Boloor et al. (2020)	○	○	○	●	●	○	O	$\mathcal{E}$	✓	✓*		✓		
	RL	Camera	Boloor et al. (2019)	○	○	○	●	●	○	O	$\mathcal{E}$	✓	✓*		✓		
	RL	Sensor	Farivar et al. (2021)	●	○	○	●	●	○	S	o		✓		✓		

Attacker's goal and perturbations. Lane keeping integrity has been targeted at training and test time. We first describe the two backdoor attacks. Both are target images (Han et al., 2022; Gong et al., 2022) and use patch triggers. Han et al. (2022) consider a physical object, traffic cones, as triggers (O), which inverts the ML model's perceived lane. BAFFLE's (Gong et al., 2022) patch trigger (P) causes the RL agent to switch to a poorly performing policy.

At test time, evasion attacks cause a wrong steering output so the AV leaves the lane (Tian et al., 2018; Sato et al., 2021; Jing et al., 2021; Song et al., 2023). Alternatively, the existing steering angle is amplified to increase accidents (Farivar et al., 2021; von Stein et al., 2023). The adversary can also change the steering output in a specific or targeted direction (Chernikova et al., 2019; Bolloor et al., 2019; Zhou et al., 2020) or make the vehicle follow a target trajectory (Bolloor et al., 2020; von Stein et al., 2023), inducing a crash (von Stein et al., 2023). These goals have been shown on both RL agents and ML models. Concerning the perturbations, analogous to the backdoor attacks, most evasion attacks rely on visual data and deploy either patches (P) (Sato et al., 2021; Jing et al., 2021), or perturbations like billboards (O) (Zhou et al., 2020; von Stein et al., 2023), shadows (O) (Bolloor et al., 2019, 2020; Song et al., 2023) or use a maneuver of another car as a perturbation (O). However, some attacks also introduce global perturbations (L_∞) (Tian et al., 2018; Chernikova et al., 2019; Sun et al., 2020) changing all pixels. An exception is introduced by Farivar et al. (2021), who affect the sensor data and thus perturb directly the numerical values (S).

Physical constraints. The latter attack by Farivar et al. (2021) assumes that the adversary is a worm within the AV's software, and changes values only during curves and within the range of benign data. The remaining papers adhere to physical changes common in vision and take into account EoT, print-ability, and varying lighting conditions (Tian et al., 2018; Zhou et al., 2020; von Stein et al., 2023; Guo et al., 2023). Han et al. (2022) use an existing object, a traffic cone, as a trigger. The DRP Attack (Sato

Table 13

End-to-end driving security. Attacks target position- or camera-based methods, and apply global (L_∞) or patch-based (P) perturbations on environment $\mathcal{E}$ or state s . Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

End-to-end driving			Attacker's knowledge			Attacker's capability										E	S
		Sensor	Reference	ω	D	f	X	Y	X_t	δ		exe.	univ.	transf.	phy. con.		
Backdoor	RL	Position	Yu and Liu (2022)	○	○	○	○	○	○	P	s		✓			○	○
	RL	Camera	Pan et al. (2022)	○	○	○	●	●	○	P	$\mathcal{E}$	✓/✓*			✓	○	○
	RL	Camera	Pan et al. (2022)	●	○	●	●	●	○	P	$\mathcal{E}$	✓*		✓	✓	○	○
	RL	Camera	Pan et al. (2022)	●	○	○	●	●	○	P	$\mathcal{E}$	✓/✓*			✓	○	○
Evasion	RL	Camera	Pan et al. (2022)	●	○	○	●	●	○	P	$\mathcal{E}$				✓	○	○
	RL	Camera	Pan et al. (2022)	●	○	○	●	●	○	P	$\mathcal{E}$				✓	○	○
	RL	Camera	Yang et al. (2020b)	○/●	○	○	●	●	○	L_∞	$\mathcal{E}$					○	○
	ML	Camera	Wu et al. (2023c)	○	○	○	●	●	○	L_∞	$\mathcal{E}$	✓	✓/			○	○

et al., 2021) uses commonly used road patches, e.g., perturbations in gray-scale that cover the road. Another possibility for realistic perturbations is using black lines similar to shadows (Boloor et al., 2019, 2020). Finally, Song et al. (2023) enforce that their perturbation-cars' maneuvers comply with existing traffic rules.

Exposure, attack severity, and qualitative risk. In addition to these constraints, four papers evaluate in a physical setting. Two test their attack on a real vehicle (Sato et al., 2021; Jing et al., 2021), and one on a production system, Baidu's Apollo (Han et al., 2022). Furthermore, the deepbillboard attack (Zhou et al., 2020) was demonstrated in the problem space. While RL-based attacks also consider physical constraints, they are not evaluated on street-legal vehicles. Attack severity polarizes, with several papers having a medium to high, and others having a low score. This leads to an on average medium risk. Many models (22) and datasets (14) support this analysis.

Conclusion. In lane keeping, although different attacks, sensors, and perturbations are investigated, the variance could be higher. There is also an absence of transferability studies, which are crucial to assessing the vulnerability of proprietary models. At the same time, there are very mature papers in terms of exposure: all covered attacks are shown at least once on a real vehicle or components thereof. Also, RL-based attacks incorporate physical constraints, yet they are seldomly evaluated in the problem space. Finally, more work on poisoning, model stealing,

4.5. End-to-end driving

Last but not least, there is also the possibility to implement autonomous driving in an end-to-end manner, e.g. using for example visual input and outputting direct steering and velocity control, without individual steps in between. We will conclude our AI security analysis by the security of these end-to-end systems.

End-to-end driving provides all required tasks within the AV in one model, and can be implemented using ML (Wu et al., 2023c) or RL (Kothari et al., 2021b). We give an overview of all found attacks in Table 13.

Attacker's goal. Yu and Liu (2022) harm integrity at training time. Their attack causes the victim to crash with other vehicles once the trigger was seen. In contrast, all evasion attacks implement an unspecific, straightforward goal for the attacker, which is reducing the victim's rewards (Yang et al., 2020b; Pan et al., 2022) or decreasing the performance (Wu et al., 2023c).

Perturbations. The trigger implemented by Yu and Liu (2022) encompasses changes to both the position and velocity of the victim AV over a range of time using the state s of the agent. All other approaches work on camera inputs, and introduce patches (P) (Pan et al., 2022) or global perturbations (L_∞) (Yang et al., 2020b; Wu et al., 2023c) of the victim's observed environment $\mathcal{E}$.

Physical constraints, exposure, attack severity and qualitative risk. Only Pan et al. (2022) apply EoT to their visual perturbations (Athalye et al., 2018). They also test their examples by printing the patches and presenting them to a real-world autonomous robot. Also Wu et al. (2023c) test on a real-world robot, where however the perception is manipulated, e.g. perturbations are not applied in the problem space. In contrast, attack severity is polarized, with some attacks by, for example, Pan et al. (2022) scoring high, whereas others obtain low scores. Combining these results would yield a low-to-medium risk, which is however supported by rather few models (8) and insufficiently many datasets (4).

Conclusion. Although this area consists of only four papers, both backdoor and evasion attacks have been studied. Despite the few papers, the sensors, paradigms, and threat models covered are rather diverse. However, more work is needed to assess end-to-end security, for example by incorporating more real-world tests, transferability studies, and targeted attacks.

5. State of research, challenges, and future research directions

Having discussed all individual tasks, we now attempt to give a big picture of the state of research in practical AV AI security. To this end, we first discuss existing research gaps and future research on the attack level. We then take a step back and discuss the AV as a system and which challenges and open questions exist on how attacks interact with this system. Finally, to assess the broader context, we discuss our findings for problem space AV AI security and the associated research questions.

Table 14

AV AI security. For each task, we list the number of papers and publication distribution (each bar represents all papers in six months, starting with Jan. to June 2017, first occurrence counts including ArXiv). A red bar marks the first paper implementing physical constraints. We denote the total amount of tested models and datasets and available models/datasets on pages like *paperswithcode*. We follow the same pattern for attacks studied and possible attacks. Analogously, we depict the number of non-studied features of all features contributing to severity and exposure. We finally plot relative risk as the mean (dark) and maximum (light) exposure and severity of all papers combined.

Section and task	# Papers and time	f	Data	Attacks	θ	Risk
4.1.1 Traffic sign Rec.	22	17/200	11/ 7	3/6+	0/9	
4.1.2 Action recognition	16	24/ 39	10/ 96	2/6+	3/9	
4.1.3 Object detection	47	61/200	22/235	4/6+	0/9	
4.1.4 Object tracking	12	11/ 17	9/ 52	2/6+	2/9	
4.1.5 Image segmentation	25	39/ 42	13/ 12	3/6+	2/9	
4.1.6 Depth estimation	6	9/ 10	2/ 22	1/6+	2/9	
4.1.7 Optical flow	8	12/ 27	5/ 12	1/6+	2/9	
4.1.8 Pose estimation	3	5/ 1	2/ 4	1/6+	3/9	
4.2.1 Trajectory prediction	8	18/ 2	8/ 28	2/6+	2/9	
4.3 Planning	12	15/ 1	9/ 1	5/6+	4/9	
4.4 Lane keeping	14	22/ 7	14/ 15	2/6+	1/9	
4.5 End-to-end	4	8/ 0	4/ 0	2/6+	2/9	

5.1. Landscape of attack evaluations

Without considering the entirety of the vehicle or practical aspects, there are research gaps that have to be tackled. These are related to the diversity of the studied tasks, attacks, and the exact evaluation considered and are discussed in this part. Before we discuss these, we remind ourselves that according to our analysis, there is a large variance in the number of papers per task. These numbers range from three in pose estimation (Section 4.1.8) to almost fifty papers in object detection (Section 4.1.3). This number greatly affects the expected research maturity for each task.

Current task coverage. Many areas, including bird's eye view analysis (Zhu et al., 2023b), graph scene generation (Zhao et al., 2024), human pose estimation (Duan et al., 2023), 3D human pose estimation (Chen et al., 2021b), and pedestrian intention prediction (Razali et al., 2021; Guo et al., 2022) are not included in our analysis as we could not find sufficiently many security analyses. Studying these areas is crucial to understanding AV's vulnerabilities that rely on these systems. In areas with several publications, as visible in Table 14, the number of attacks studied per task varies. Evasion is studied in all fields and is the most studied attack overall: 138 of 177 papers, or 78%, are about evasion. While attacks via the test data may seem more practical on proprietary, safety-relevant products, other test-time attacks like sponge attacks are less frequently studied. Instead, the second most frequent attack with 29 of 177 papers, or 16.4%, are backdoors. Backdoors are studied in almost all fields except depth estimation (Section 4.1.6), camera pose estimation (Section 4.1.8), and optical flow (Section 4.1.7). In all three fields, evasion is the only attack studied; vulnerability to other threats is currently unknown. For an in-depth understanding of AV AI security, insights about all attacks on all tasks are needed.

Current attack diversity. The fields with the broadest security assessment are image classification (Section 4.1.1), object detection (Section 4.1.3), image segmentation (Section 4.1.5), and planning (Section 4.3), where all here discussed attacks are studied, however sometimes outside the here considered setting of AVs. In all other areas, some attacks are left out, particularly sloth, privacy attacks and model stealing in the AV context. Orthogonally, it is crucial to study a variety of attack variations. For example in pose estimation, all three papers examine the same evasion black-box threat model (Section 4.1.8). In end-to-end driving (Section 4.5) and image segmentation (Section 4.1.5), only one kind of backdoor attack is studied. This lack of variety may be due to few publications—areas with several publications tend to cover diverse attack kinds, as evident in tasks like trajectory prediction (Section 4.2.1) or planning (Section 4.3). Sheatsley et al. (2023) show the importance of such a variety within attack evaluations. Although they focus on white-box attacks, they show that some vision datasets (e.g., CIFAR10) and perturbation measurements do not generalize. Similar studies are also required for threat models in other areas and tasks, like the ones discussed here, to understand the effect and importance of different threat model properties for each task.

Attack evaluations. This latter work by Sheatsley et al. (2023) raises the question of whether security tests on data sets can be reliable and generalize across datasets. However drawing from existing work in image classification, we must also consider failure cases in attack evaluations. As discussed in the background, most attacks are based directly on the model's gradient, which can be unavailable or obfuscated due to unstable predictions (Pintor et al., 2022). Other, attack-specific failures indicated by Pintor et al. (2022) are non-convergence due to misconfigured hyperparameters or a failure to transfer from an existing attack due to the wrong attack loss. An unadapted loss from an existing attack can also lead the attack to a bad local optimum and thus prevent the attack from succeeding. Pintor et al. (2022) finally remark that other implementation errors may also lead to a wrong security assessment. In particular, when proposing defenses, this assessment becomes harder (Shen et al., 2022), as defenses have their own complex set of evaluation design requirements (Carlini et al., 2019; Cinà et al., 2023).

Summary and Future work. This subsection identified future work required for individual attacks. This entailed areas like bird eye view, graph scene generation, or pedestrian intention prediction which are currently missing, and areas like optical flow or end-to-end prediction, where more attacks need to be studied. Some attacks, like privacy threats or model stealing, should be researched with a focus on AVs. In addition, we summarized the need for good threat modeling and thorough evaluations taking into account whether attacks succeed or fail.

5.2. AV system security evaluations

Beyond challenges at the attack level, AVs have to be studied as a system made from (potentially) various AI components. This view brings new challenges: Firstly, different sensors, approaches, and paradigms must be evaluated. Furthermore, all parts or components must be studied in relation to each other, not in isolation. Finally, the system may use redundant sensors or components, further affecting security.

Sensors, approaches, and paradigms. An AV as a system contains different sensors and may implement different approaches or paradigms for a task. While all of these need to be security tested to ensure robustness to tampering, the found assessment is not balanced. For example, there is an overall bias towards camera-based approaches in object detection (Section 4.1.3), object tracking (Section 4.1.4), lane keeping (Section 4.4), and end-to-end driving (Section 4.5), although other sensor types such as LiDAR or position-based approaches could be used. In general, when several ways to solve a task exist, these are not necessarily studied with the same rigor. In, for example, image segmentation (Section 4.1.5), more work focuses on semantic segmentation; than on instance segmentation. Analogously, most works study security on coordinate-based trajectory prediction (Section 4.2.1), not map-based approaches. Finally, in planning (Section 4.3), most attacks target state-based approaches. Paradigms are also not always balanced. In lane keeping (Section 4.4), nine of fourteen papers focus on ML-based approaches vs RL-based, in end-to-end driving, more works rely on RL (6 RL papers vs. 1 ML paper). Overall, RL is not as well studied as ML, even when considering papers outside the AV context (Demontis et al., 2022). While one may argue that always the best-studied component can be used, sensors, approaches, and paradigms may have security advantages and disadvantages, which we can benefit of if they are well understood.

AVs as AI systems. When considering the AV as a system, not a set of tasks, previous work found downstream AV components were studied less (Shen et al., 2022). In contrast, we found high and middle-risk applications in perception and high-risk tasks in control (Table 14). We do acknowledge, however, that certain downstream tasks like action recognition (Section 4.1.2) and planning (Section 4.3) are less mature than object detection (Table 14), for example. In mature areas like traffic sign recognition (Section 4.1.1), object detection (Section 4.1.3) or lane keeping (Section 4.4), subsystems have been shown to fail, and their failure triggered the AV to perform dangerous actions such as leaving the lane (Nassi et al., 2019; Morgulis et al., 2019; Cao et al., 2021; Jing et al., 2021; Hallyburton et al., 2022). Beyond these few cases, there is little work that investigates whether attacks propagate within an AV. This includes Cao et al. (2022), who show that their attack on trajectory prediction can affect the following planning step. Koren et al. (2022) evaluates how their attack on optical flow systems increases the error in downstream time-to-collision computations. More work is however needed to understand the effect of attacks on components, how this affects potential downstream tasks, and thus the impact of an attack on the AV. To this end, also knowledge about real-world AVs and their components is required.

Security across modalities and tasks. One way to limit an attacker's possible effect on a system could be to introduce redundancy. However, existing works show that, for example, object detection and image classification (Lovisotto et al., 2021) or depth estimation and image segmentation (Zhang et al., 2020b) are vulnerable to the same attacks. Agnihotri and Keuper (2023) extend this finding to more pixel-based prediction attacks, including semantic segmentation, depth estimation, or optical flow. Also, less related tasks can be targeted jointly, as Chawla et al. (2022) show for pose and depth estimation. Additional work shows that even algorithms relying on different sensors can be targeted. For example, Abdelfattah et al. (2021) show that an object detector relying on LiDAR and a camera can be fooled when both representations are simultaneously attacked. Cao et al. (2021) confirm these results by printing their perturbations to test in the problem space, and show success also in a simulation. It is thus crucial to understand which components are used together in a vehicle, and which are redundant or could be implemented redundantly. Moreover, there is a gap in terms of analyzing a system in simulation and the real-world (Stocco et al., 2022), highlighting one more the need to study street-legal vehicles.

Summary and future work. This subsection identified the need to balance the security analysis of different sensors, AI approaches, and paradigms. The underlying rationale is that research should provide a full picture of possible applied methods and their interactions to assess attack impact on individual components and the entire AV. To streamline needed resources, research should be based as closely as possible on existing systems to assess their vulnerability. Another possibility is high-level insights, for example, studying attack flow through complex systems or across tasks.

5.3. Real-world or problem-space security evaluations

Having discussed AV AI security at the attack and system level, we now focus on the real-world risk assessment. In other words, understanding which problem space attacks exist on street-legal vehicles. To this end, we first review which specific tests are required to come closer to such an assessment, before we discuss required resources. We conclude the section by discussing the role of transferability in real-world studies. Before we start, we briefly discuss the occurrence of real-world aspects within each task. Some areas' security has been studied since 2017 (for example, image segmentation), and physical constraints (red bars in Table 14)

were considered in the earliest papers. In contrast, action recognition (Section 4.1.2) has been studied since 2018, yet no paper implements physical constraints so far.

Current landscape of real-world attacks. Our measures exposure and severity rely on nine features described in the methodology (Section 3), including, for example, tests on production components or real vehicles as well as black box access, transferability studies, and universal attacks, among others. In Table 14, we denote with $\emptyset$ the amount of these nine features absent or unstudied within one task. For most tasks, there are no tests on real-world vehicles or production components; which are only used in traffic sign recognition (Section 4.1.1), object detection (Section 4.1.3) and lane keeping (Section 4.4). However, some areas are short in other of these practical aspects. For example, within action recognition, no paper considers physical constraints (Section 4.1.2). There are no real-world evaluations in planning (Section 4.3). Alongside lane keeping (Section 4.4), there are also no experiments on transferability in planning. In addition, there is an absence of black-box attacks in pose estimation (Section 4.1.8). However, studies on street-legal vehicles and production components or black-box and transferability experiments are crucial to understanding real-world, proprietary system vulnerabilities. This extends to how vulnerable these systems are when attacked via their sensors and input data, and should be investigated in depth.

Current landscape of attack resources. We have, so far, ignored important aspects of real-world vulnerabilities. For example, the amount of data required for an attack, or other resources like compute. In the case of evasion, around one third (44) of the articles did not specify any runtime at all, most instead reported iterations or epochs. Yet, this information is crucial, as the computing time ranges from less than a second (Liu et al., 2019) to several hours (Chow et al., 2020). Few works, like for example Huang et al. (2023a), explicitly focus on crafting runtime and comparing several attacks, concluding that longer runtime leads to a higher success rate. Other attacks, for example, backdoors, generally report attack parameters like the percentage of poisoned training samples. However, understanding which fraction of data can potentially be affected by an attacker in an AV context is still an open research question. In this sense, we need a better understanding of the available and required resources for the attacker and which resources are potentially accessible in real-world vehicles and their manufacturers.

Transferability. In all cases where knowledge is absent, transferability is crucial for real-world vulnerabilities, as it allows us to study a model that is proprietary and not accessible. However, almost all papers consider model-to-model transferability, implicitly assuming knowledge about the training data. As data may be proprietary, we join calls by Gupta et al. (2022) and Sheatsley et al. (2023) to study data-to-data transferability, where the data is also unknown. Transferability is also important in other contexts: when multiple tasks can be fooled at once, this implies that overall less knowledge is used about the implemented task in the AV at hand (Zhang et al., 2020b; Agnihotri and Keuper, 2023). However, transferability may itself be affected by problem space parameters. Lovisotto et al. (2021) showed, for example, the importance of lighting in transferability, where lower light settings lead to better transferability. More research is also needed to understand the effect of specific parameters, like the scaling parameter in image segmentation (Arnab et al., 2018; Gu et al., 2021), or the effect of different representations like coordinate and map based in trajectory prediction (Section 4.2.1). Currently, it is also unknown whether transferability holds when a task is implemented in RL or across different paradigms such as ML and RL.

Summary and future work. This subsection discussed and identified necessary research to obtain necessary knowledge about real-world vulnerabilities in AVs. While one part consists of understanding which knowledge and resources are available to the attacker, it is also crucial to study real-world scenarios via tests on vehicles, production systems, or at least conduct black-box attacks or transferability studies.

6. Implications beyond research

While the research to understand the underlying, full AV attack surface and AV component vulnerabilities is necessary, the implications of our analysis extend beyond research. As stated in the introduction, attacks may render AVs unresponsive, tamper with their perception, and affect how the AV understands other agents around it. Such attacks influence the passengers' comfort and may cause material damage or fatalities. Our work has thus concrete implications for both the automotive industry and related policies, which we summarize in this section.

In addition, in the worst case, AVs under attack can cause physical harm or death to passengers or others: testing AVs in the real world has already led to fatalities.⁵ Constant, real-world tests have furthermore caused resentments towards the AVs in some cities.⁶ From an ethical perspective, there is thus the duty jointly to researchers, manufacturers and policymakers to limit negative impacts as much as possible. These include, among others, how to disclose attacks responsibly, how to avoid attack code being easily accessible by attackers, and how to security test vehicles in the real world without endangering anyone.

6.1. Implications for the automotive industry

Even beyond immediate security concerns, robustness is related to benign performance and may improve existing systems. Examples include Bahari et al. (2022), who search for physically possible failure cases and find existing intersections and road scenarios where the model will fail (Section 4.2.1). Analogously, Rempe et al. (2022) and Hanselmann et al. (2022) find driving patterns that cause collisions (Section 4.3). Other examples include misclassification due to rotation of objects (Dong et al., 2022)

⁵ <https://www.washingtonpost.com/technology/2023/06/10/tesla-autopilot-crashes-elon-musk/>.

⁶ <https://www.trafficechnologytoday.com/videos/video-angry-mob-attacks-autonomous-vehicle-in-san-francisco.html>.

(Section 4.1.1), safety-relevant trajectories (Ren et al., 2023), or an agent's malicious positions in planning (Vemprala and Kapoor, 2021) (Section 4.3). It may thus be beneficial for practitioners to be aware of the existing working AI security.

Secondly, and as reasoned in the previous Section (Section 5.2), researchers need access to real-world vehicles to understand not only the security of individual systems digitally but how attacks affect several components and propagate through the AV (Shen et al., 2022). As previously stated, it is crucial to test the street-legal vehicle, as tests in simulations may not be faithful (Stocco et al., 2022), or security assessments may not generalize to new settings (Sheatsley et al., 2023). However, only the developing company may grant full access to their AVs. While concerns about intellectual properties and publications of results may be challenging, such access is required to build secure vehicles. Initiating and fostering such collaborations would thus be highly beneficial.

Lastly, and probably most straightforward, our analysis showed exploits on street legal cars (Sections 4.1.1, 4.1.3 and 4.4). Manufacturers should thus take into account security concerns when building AVs. This could be by building redundant systems considering shown vulnerabilities affecting multiple components (Section 5.2). Orthogonally, better-understood components could be used (Section 5.1). Also, blue and red teaming may support security analysis (Mirkovic et al., 2008). Here, one team (red) attacks the model or system, while the other team (blue) defends or mitigates the attack. However, both previous approaches require good threat modeling and knowledge about attacks that are, at least in theory, possible.

6.2. Implications for policymakers

Our work provides an overview of AVs' AI vulnerabilities, outlining existing and potential threats (Section 4). Given that AVs are already or will become part of critical infrastructure, understanding associated risks is crucial for upcoming legislation and public response to security-related events. More concretely, legislation should acknowledge the existence of tampering with AI-based vehicles and allow the conviction of attackers. Orthogonally, executive forces should be educated and able to deal with such cases. For example, traffic police should be expanded by intelligence and personnel in both AI and security.

Yet, as stated in Section 5.3, it is crucial to gain an understanding of real-world attacks. Such AI security incidents should thus be collected. While each company may document occurred attacks individually, collecting incidents related to AI (security) in AVs and over the entire industry may be more efficient. While there are databases to collect incidents already, they are either small and self-reported (Grosse et al., 2024b) or do not focus on AVs or AI security but on AI incidents in general (McGregor, 2021).⁷ While existing legislation like, for example, the EU AI Act requires the collection of incidents,⁸ these cover again AI incidents in general, not specific security incidents. We suggest that AI security incidents should be collected centrally and enable an in-depth analysis, analogous to incident reporting in aviation or the CVE program.⁹ Such a database would also enable detecting increases in attacks and enable appropriate responses.

7. Concluding remarks

In this review, we analyzed qualitative real-world security risks of AI in AVs. Of twelve areas, three demonstrated problem-space attacks on street-legal cars. AVs are thus already vulnerable to attacks on their AI components. However, existing security analyses of AVs are not balanced in terms of attacks, tasks, sensors, AI approaches, and AI paradigms. Furthermore, AVs pose specific challenges as they have to be studied as a system, not a single component. These challenges exist on top of the difficulty of proper evaluations in AI security. We also outlined implications for the automotive industry in terms of existing benefits and risks associated with security and needed collaboration. Finally, we summarized the implications for policymakers like the need for incident collection.

CRedit authorship contribution statement

Kathrin Grosse: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Conceptualization. **Alexandre Alahi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition, Conceptualization.

Acknowledgments

The authors thank the VITA-lab for the helpful discussions and support. Special thanks go to Kaouther Messaoud Ben Amor, Charles Corbière, George Adaimi, Vladimir Somers, Reyhaneh Hosseinejad, and Taylor Mordan. The authors would also like to thank the anonymous reviewers for their helpful feedback.

Appendix. Object detection overview table

See Table A.15.

⁷ <https://avidml.org/database/>.

⁸ <https://cset.georgetown.edu/article/the-eu-ai-act-a-primer/>.

⁹ <https://cve.mitre.org/>.

Table A.15

Object detection security. Methods are mainly based on visual cameras or LiDAR and attempt to classify vehicles (V), pedestrians (P) or traffic signs (S). Perturbations are global (L_∞), patch-(P) or object shaped (O) and executed in two or three dimensions. We further denote Targeted, Sequential, Universal and Physical Constraints. Threat model and attack properties as in Section 2.2 and exposure E and severity S as in Section 3.

					Attacker's knowledge		Attacker's capability																	
	Sensor	V	P	TS		Reference	ω	D	f	X	Y	X_t	δ		T	Exe.	U	Tr	PC	E	S			
Backdoor	LiDAR	✓	✓		Backdoor points (Xiang et al., 2021)	●	○	●	●	○	○	P	3D	/T	✓	✓		✓			○	○		
	Camera			✓	Badnets (Gu et al., 2017)	○	○	○	○	○	○	P	T	✓	✓					○	○			
	Camera	✓	✓	✓	Tpatch (Zhu et al., 2023a)	○/●	○	○/●	○	○	○	P/ L_∞	/H/C	✓	✓	✓	✓			○	○			
	Camera	✓			MBA (Wei et al., 2023)	●	○	●	○	○	○	O	H	✓	✓					○	○			
	Camera	✓	✓	✓	BadDet (Chan et al., 2022)	●	○	●	●	○	○	P	/H	✓	✓		✓			○	○			
	Camera	✓	✓	✓	Luo et al. (2023)	●	○	●	●	○	○	P	H	✓	✓		✓			○	○			
	Camera	✓			Claoking B. Ma et al. (2022)	●	○	●	●	○	○	O	3D	H	✓*	✓		✓		○	○			
	Camera	✓			Wu et al. (2023a)	●	○	●	○	●	○	P	H	✓	✓	✓				○	○			
Evasion	GPS	✓	✓	✓	FLAT (Li et al., 2021b)	○/●	○	○/●	●	●	○	L_∞			✓			✓	✓		○	○		
	LiDAR		✓		Cai et al. (2020)	○	○	○	●	●	○	P	3D	H							○	○		
	LiDAR	✓	✓	✓	ISO (Wicker and Kwiatkowska, 2019)	○/●	○	○	○	●	●	○	P	3D							○	○		
	LiDAR		✓		Adv-LiDAR (Cao et al., 2019)	●	○	●	●	●	○	O	3D		✓				✓		○	○		
	LiDAR		✓		Tu et al. (2020)	○/●	○	○/●	●	●	○	O	3D	H			✓	✓	✓		○	○		
	LiDAR		✓		Wang et al. (2021a)	○/●	○	○/●	●	●	○	P	3D	H				✓			○	○		
	LiDAR	✓	✓		Jin et al. (2023)	○/●	○	○/●	●	●	○	O	3D	H/C	✓	✓			✓		○	○		
	LiDAR	✓	✓	✓	PRA (Cao et al., 2023a)	●	●	●	●	●	●	P	3D	H							○	○		
	Camera	✓	✓	✓	UEA (Wei et al., 2019a)	○	○	○	●	●	○	O/ L_∞	H						✓		○	○		
	Camera	✓			Adv. T-shirt (Xu et al., 2020)	○	○	○	○	●	●	○	P	3D	H	✓	✓			✓		○	○	
	Camera	✓			Thys et al. (2019)	○	○	○	●	●	○	P	H	✓	✓	✓			✓		○	○		
	Camera		✓		DTA (Suryanto et al., 2022)	○	○	○	●	●	○	O	3D		✓*	✓			✓		○	○		
	Camera		✓		Jia et al. (2020)	○	○	○	●	●	○	P	T								○	○		
	Camera		✓		NaturalAE (Xue et al., 2021)	○	○	○	●	●	○	O/ L_∞		✓*	✓	✓	✓	✓			○	○		
	Camera		✓		GhostImage (Man et al., 2020)	○	○	○	●	●	○	P	/C						✓		○	○		
	Camera		✓		Zeng et al. (2019)	○/●	○	○	○	●	●	○	O			✓	✓			✓		○	○	
	Camera	✓	✓	✓	Poltergeist (Ji et al., 2021)	●	○	○	○	●	●	○	L_∞	/H/C	✓*	✓	✓		✓		○	○		
	Camera		✓		CCA (Hu et al., 2021)	●	○	○	○	●	●	○	O	3D	H	✓*	✓			✓		○	○	
	Camera		✓		Camou (Zhang et al., 2019)	●	○	○	○	●	●	○	O	3D	H	✓*	✓					○	○	
	Camera		✓		Wu et al. (2020b)	●	○	○	○	●	●	○	O	3D	H	✓*	✓			✓		○	○	
	Camera		✓		Adv3D (Li et al., 2023)	●	○	○	○	●	●	○	O	3D	H	✓	✓	✓	✓	✓		○	○	
	Camera	✓	✓	✓	Kuang et al. (2021)	●	○	●	○	●	●	○	L_∞	T							○	○		
	Camera	✓	✓	✓	TOG (Chow et al., 2020)	○/●	○	○/●	●	●	○	P/ L_∞	/H	✓	✓	✓	✓					○	○	
	Camera	✓	✓	✓	Zolfi et al. (2021)	○/●	○	○/●	●	●	○	P	T	✓*	✓	✓	✓	✓	✓			○	○	
	Camera	✓			Wu et al. (2020a)	○/●	○	○/●	●	●	○	○	3D	H	✓*	✓	✓	✓				○	○	
	Camera		✓		UPC (Huang et al., 2020)	○/●	○	○/●	●	●	○	O/ L_∞	3D		✓*	✓	✓	✓	✓			○	○	
	Camera		✓		DAS (Wang et al., 2021b)	○/●	○	○/●	●	●	○	O	3D		✓*	✓	✓	✓	✓			○	○	
	Camera		✓		RAP (Li et al., 2018)	○/●	○	○/●	●	●	○	L_∞	H						✓			○	○	
	Camera		✓		Lu et al. (2017)	○/●	○	○/●	●	●	○	O	H	✓*	✓	✓	✓					○	○	
	Camera		✓		ShapeShifter (Chen et al., 2018)	○/●	○	○/●	●	●	○	O	/T	✓*	✓	✓	✓	✓	✓			○	○	
	Camera		✓		Zhao et al. (2019)	○/●	○	○/●	●	●	○	P	H/C	✓*	✓	✓	✓	✓	✓			○	○	
	Camera		✓		SLAP (Lovisotto et al., 2021)	○/●	○	○/●	●	●	○	O	T	✓*	✓		✓	✓				○	○	
	Fusion	✓	✓	✓	MSF-ADV (Cao et al., 2021)	○	○	○	○	●	●	○	O	3D	H	✓*			✓	✓		○	○	
	Fusion		✓		Abdelfattah et al. (2021)	○	○	○	○	●	●	○	O	3D		✓	✓			✓		○	○	
Fusion		✓		Frustum Att. Hallyburton et al. (2022)	●	○	●	○	●	●	○	P	3D	T/C	✓/	✓			✓		○	○		
Fusion	✓		✓	Nassi et al. (2020)	●	●	●	●	●	●	○	O		✓*	✓			✓			○	○		
M. stealing	Camera	✓	✓	✓	Liang et al. (2022)	●	○	○/●	●	●	○		f								○	○		
Sponge	LiDAR				Liu et al. (2023b)	○	○	○	●	●	○	P	3D					✓			○	○		
	Camera	✓	✓		Shapira et al. (2023)	○	○	○	●	●	○	L_∞		✓	✓						○	○		

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